Score-linked practice intelligence from long-form violin training video

Alan N. Pham[1]

[1]AO Labs, Worcester, MA, USA

High-level music training produces dense evidence, but daily practice videos rarely become a usable longitudinal record. We report Curtis, a live AO Labs system that converts long-form violin practice media into source-linked practice records: active-playing windows, detected audio fragments, repertoire labels, score-alignment candidates, source-score review queues, and accepted/rejected evidence gates. At the 15 May 2026 evidence lock, the system indexed the public `@nalalan` YouTube channel, separated uploaded-video duration from detected active playing, and required explicit agreement between audio, transcription, parsed source score, visible source crop, and musician review before showing any score-linked phrase as accepted evidence. The current strict state intentionally returns the rejected Scherzo-Tarantelle phrase lane to zero accepted score-linked phrases. The contribution is not an admissions prediction or a solved transcription benchmark. It is a falsifiable method for turning messy practice video into inspectable musical evidence while preserving uncertainty, rejecting wrong matches, and making future practice, score matching, and feedback claims auditable.

1 Introduction

Elite music preparation depends on repeated, deliberate practice and feedback over long time horizons (*1–3*). The useful data are not only polished performances. They include ordinary practice sessions, failed takes, camera angle limitations, recurring technical problems, and changes in focus across days. A violinist recording daily practice can therefore produce a large longitudinal dataset, but the raw dataset does not automatically become an operating record.

Curtis addresses this problem as a live software system rather than as a portfolio page (*4*). The system indexes public practice media, separates uploaded video duration from detected violin-playing practice time, records technical observations, and keeps evidence tied to clips and generated notation. It is designed for the Curtis Institute goal, but it does not claim to estimate an acceptance probability. The current role of the system is narrower: observe practice evidence, retain history, separate judged from unjudged evidence, and identify the specific passage-level blockers that can be tested in subsequent recordings.

The core pipeline is video → active practice detection → clips → music transcription → repeat grouping → score matching → heat map → repertoire update → passage-level observations. It is built against established difficulties in onset detection, pitch estimation, automatic music transcription, and music information retrieval rather than assuming that a video can be cleanly converted into a score (*5–9*). The output is intentionally small: analyzed practice days and an automatically updated repertoire list, both grounded in clips, notation, active playing time, and confidence limits.

The system is also part of AO Labs' broader progress architecture. In May 2026, AO Progress was introduced as a shared ledger for cross-application state and long-term changes across public apps, papers, CV artifacts, Curtis practice state, Imagineer state, and Relay state (*10*). Curtis supplies one stream in that ledger: timestamped media and review evidence for musical development.

2 System state

The production state uses the live backend at `https://curtis.aolabs.io/api/curtis/media-status`, public YouTube metadata, MusicXML-derived score parsing, and local source-score assets where available (*11–13*). The checked source state at 2026-05-15 05:30 PM EDT reported the values in Table 1. These are operating facts, not outcome claims.

A later implementation check at 2026-05-14 08:29 AM EDT corrected and extended the Scherzo-Tarantelle score-truth source, but subsequent review showed that the intermediate `D C Bb` opening map was still a staff-position error rather than accepted source truth. The previous `D A# G D` source-score phrase, the later `D C Bb D C Bb D` source map, and the five-note `A#4 D5 C5 A#4 D5` / `Bb4 D5 C5 Bb4 D5` display lane are all rejected as accepted symbolic-score evidence. The 2026-05-14 07:04 PM EDT actual-source score-gate update added an exact-range source crop for that five-note subrange and prohibited generated notation SVGs from appearing as accepted score snippets. The 2026-05-14 09:45 PM EDT spelling-context update closed a separate visual mismatch by carrying the source key signature into the transcription panel and requiring visual-range and score-spelling agreement. The 2026-05-14 10:52 PM EDT visual-crop demotion then rejected the same source crop because Alan's review showed that the visible source-score notes still did not match the transcription. Generated notation remains available only as the transcription-side rendering; a score panel must use a non-generated source-score image whose visible noteheads, register, spelling context, parsed score-note range, and explicit visible-note-sequence verification all match the transcription. The 2026-05-15 05:00 PM EDT strict evidence-gate update makes that demotion enforceable in code: a score crop cannot count as visible accepted evidence unless score box centers, visible score notes, written spelling, exact note-plus-octave sequence, audio-to-transcription agreement, transcription-to-score agreement, and an accepted truth-evidence flag all pass. The 2026-05-14 02:51 PM EDT target-triage update checks four-note-or-longer reference-audio targets against the current verified symbolic score excerpt before any promotion attempt. In that strict-gate state, the top source target was `D D D# D D A# G`; under the latest corrected source map its best exact-MIDI overlap is the single `G5`, so it remains not accepted score evidence. The 2026-05-14 03:49 PM EDT score-map workbench update added a local source-map verification queue with 779 unaccepted score glyph candidates across 7 staves from page 2 of the local IMSLP solo part. The 2026-05-14 04:24 PM EDT note-hypothesis update filters that queue into 172 unaccepted likely notehead hypotheses with staff-position pitch labels. The 2026-05-14 05:17 PM EDT review-packet update fixed the skipped-staff failure mode in the source detector, expands page-2 coverage to 951 unaccepted score glyphs and 214 unaccepted note hypotheses across 8 staves, and writes 8 staff-level source review packets. The 2026-05-15 12:45 AM EDT truth-workbench update adds a persistent accepted/rejected/pending truth-item path and a compact site surface for queued source targets, accepted truth clips, rejected mismatches, and regression benchmarks; it also makes the visible-note-sequence flag mandatory for score-visual and actual-source score locks. Before the strict-source correction, the deployed strict-gate tracker reported 43.126% exact weighted progress, with 0 accepted long phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 score-coordinate heat-map fragments, 0 score-visual locks, 0 actual-source score locks, 0 accepted truth items, 2 source-verification targets queued for truth review, 14 pitch-sequence groups, 2 source-target score checks, 0 verified source targets, 7 verified symbolic score notes, 0 verified source snippets, 951 unaccepted score glyph candidates, 214 unaccepted score-note hypotheses, and 8 staff review packets. That was implementation progress only; at that point a five-note candidate, rejected source-target checks, queued truth checks, and unaccepted score-glyph, note-hypothesis, or review-packet candidates did not solve full long-phrase or full-session transcription.

The 2026-05-15 06:34 PM EDT strict-source correction temporarily promoted one narrow 2026-05-03 Scherzo-Tarantelle lane: detected notes `A#4 D5 C5 A#4 D5`, displayed as `Bb4 D5 C5 Bb4 D5`, matched to an octave-aware score sequence, and backed by a non-generated source-score crop from the local Scherzo-Tarantelle solo part. Alan's later review rejected that five-note phrase as still incorrect. Curtis now treats the 06:34 PM state as a superseded attempt, not accepted evidence. The current strict state has 0 accepted score-linked phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 score-coordinate heat-map fragments, and 0 verified source snippets from that lane.

The 2026-05-15 08:37 PM EDT fast-transition candidate update addresses the next technical blocker directly. The broader 2026-05-03 Scherzo-Tarantelle audio stream is still not trustworthy as notation: the pYIN/onset path sees many repeated D/B-flat/G events and misses enough fast motion that no longer exact score phrase can be promoted without faking evidence. Curtis now stores a hidden YIN frame-transition trace as `scoreMatchCandidateNotes`. These candidates are used only as search input for the strict score matcher. They do not render as accepted notation, do not change practice time, and cannot promote a score snippet unless the same ordered detected MIDI notes match parsed source-score MIDI notes, the written score spelling and crop agree, local audio/video are present, and no matched detected note is marked uncertain by octave correction. A local check on the 5/3 `Njh8_zq9_DM-9465` clip produced 174 hidden candidate notes with 8 pitch classes. After the later five-note rejection, those candidates remain search data only; no visible result is accepted.

The 2026-05-15 08:48 PM EDT notation-renderer correction moves the SVG treble clef baseline upward and re-anchors flat key-signature glyphs so B-flat and E-flat appear on the intended staff positions. This is a display correction only; it does not promote any transcription or score match without the stricter audio/source-score gate.

The 2026-05-15 09:07 PM EDT targeted-transcription update adds a direct reprocess path for specific media samples through `/api/curtis/transcribe/run?sampleId=...`. This reduces the burden of waiting through broad eight-sample queue passes when a known evidence clip needs to be regenerated under a new transcription pipeline. It is an implementation-speed improvement, not an acceptance shortcut: targeted runs still use the same strict hidden-candidate and source-score gates.

The 2026-05-15 09:20 PM EDT live targeted-reprocess verification used that route on `Njh8_zq9_DM-9465`. The deployed API then reported 67.375% exact weighted progress, 1 accepted long-phrase counter item, 1 accepted measure-match counter item, 1 exact-source visible score group for 2026-05-03, and 1380 hidden transition-trace candidates across the stored transcription results. That public state is now superseded: Alan reviewed the displayed `Bb4 D5 C5 Bb4 D5` phrase and rejected it. The useful retained result is the targeted-reprocess route and hidden-candidate search path, not the promoted phrase.

The 2026-05-15 09:24 PM EDT post-deploy check showed the live scanner had processed additional stored transcription samples after deployment: the live API reported 68.125% exact weighted progress and 2476 hidden transition-trace candidates. That visible one-group state is now rejected rather than accepted. The current correction requires the live counter to return to 0 accepted long phrases and 0 accepted measure matches until a new phrase survives review.

The 2026-05-15 10:05 PM EDT five-note rejection lock is the current acceptance state. Alan rejected the displayed five-note Scherzo-Tarantelle phrase itself, so Curtis now demotes `A#4 D5 C5 A#4 D5` / `Bb4 D5 C5 Bb4 D5` from accepted evidence back to candidate-only material. Regression tests require this exact lane to have zero verified score location, zero verified source snippet, no accepted measure, no accepted long phrase, no score-coordinate heat-map fragment, and no visible accepted score panel. The only note detections Alan had described as useful remain short audio-only D/A detections; the A4/A5 score anchors, D–B-flat–G–D score sequence, and five-note Scherzo phrase are rejected until new independent review proves otherwise.

The 2026-05-15 10:54 PM EDT second-pass audio-gate update targets the loop that caused repeated false progress. The hidden fast-transition trace could generate plausible note sequences before each note had independent audio confirmation, and reference-audio pitch traces could create short overlaps that looked like matches without being usable phrase evidence. A local audit of the rejected 2026-05-03 `Njh8_zq9_DM-9465` clip read the candidate closer to `Bb4 D5 Bb4 D5`; the displayed middle `C5` did not survive the second pass. Curtis now preserves audio-agreement provenance in daily detected-note series and filters hidden `scoreMatchCandidateNotes` before score search unless every note has an independent agreement source, spectral/onset participation, adequate confidence, and adequate duration. Symbolic score matches and reference-sequence matches now use the same per-note audio gate. Reference-audio pitch-trace matches require at least four notes and three distinct pitch classes, so one- or two-note coincidences do not count as phrase progress. Regression tests cover the rejected Scherzo phrase with a weak `C5`, unagreed fast candidate notes, uncertain octave-corrected candidates, and weak reference-audio matches. This update does not create an accepted long phrase; it removes a false-positive lane before the next phrase attempt.

The 2026-05-15 11:17 PM EDT candidate-isolation update closes the remaining daily-record leak. Source-verification candidates and reference-audio pitch-sequence groups are now separated from displayable score matches. Candidates may populate `candidateMatchGroups`, the truth workbench, and source-verification target queues, but the user-facing `matchGroups` bucket stays empty until exact source-score crop, visible note sequence, box centers, paired audio, transcription, score spelling, and accepted truth-evidence flags all pass. A regression test covers a playable Wieniawski source-sequence candidate that remains queued without becoming a visible match, score link, reference link, or heat-map fragment.

The 2026-05-15 11:32 PM EDT live verification reported the strict post-update state: 46.125% exact weighted implementation progress, 0 accepted long phrases, 0 accepted measures, 0 exact score-aligned windows, 0 actual-source score snippets, and 0 score-coordinate heat-map fragments. The 2026-05-03 daily record had 0 visible `matchGroups`, 6 queued `candidateMatchGroups`, and no score or reference link. The current top queued target is `A D D F# D`, which remains rejected as accepted score evidence because it does not form a contiguous exact-MIDI run in the corrected source map.

The 2026-05-16 source-position and exact-MIDI update closes the next source-map error. Re-rendering the local IMSLP solo part at high resolution and cross-checking the Wieniawski Society bars 5–9 example showed that the opening violin entrance is `A5 G5 F5 A5 G5 F5 A5 G#5 F5`, not the prior `D6 C6 Bb5 D6 C6 Bb5 D6 C#6 B5` map. The source MusicXML now stores those nine actual source notes from measures 5–7, and the older lower-octave and wrong-staff-position maps are rejected by regression tests. Curtis also removes the remaining pitch-letter-only loophole from source-verification targets. A longer target may still enter the queue from pitch-class search, but the score check now fails closed unless the detected notes carry exact MIDI values, every detected note passes paired-audio agreement, and the detected MIDI sequence appears contiguously in the parsed source MusicXML MIDI sequence. The new positive regression target is `G5 F5 A5 G#5 F5`, whose detected MIDI sequence `79 77 81 80 77` matches the same contiguous source MIDI sequence exactly. This does not promote any current live Scherzo-Tarantelle phrase; it prevents `A` matching a wrong score note, a wrong register, or a staff-position mistake from being reported as score evidence.

Table 1. Curtis source state checked at 2026-05-15 05:30 PM EDT.

Signal	Current value
Primary source	YouTube channel `@nalalan`
Inventory	219 indexed videos
Practice candidates	112 videos
Long-form candidates	92 videos
Uploaded-video marker	`violin 1`, uploaded 20 December 2025
Uploaded-video ledger	48 videos, 234 h 57 min uploaded session duration
Estimated total practice time	223 h 5 min extrapolated from the checked-window active-playing ratio; provisional because the checked windows remain sparse and sample-biased
Measured practice time	2 h 21 min of detected violin-playing footage; this is independent from transcription and cannot exceed checked video coverage
Video checked for practice time	2 h 28 min of the 234 h 57 min uploaded-video ledger
Active-practice scan	331 active sub-intervals persisted from stored media samples; 54 min 2 s of scan-derived active audio; 106 sample results; `violin 1` windows through `*2160-2250` checked; 250 queued windows still pending owner media
Unchecked archive duration	232 h 28 min still awaiting violin-playing detection
Media samples	106 active-scan sample results; 94 active violin; 12 checked non-violin
Reviewed sections	80 sections
Curtis-focused findings	80 findings
Latest dated practice log	5-6-26
Review model	GPT-5.5 for vision/text, separate audio models for audio evidence and piece verification
Confirmed labels	5/1 Haydn Symphony No. 94 IV; 5/2 and 5/3 Wieniawski Scherzo-Tarantelle
Training sources	3 source-confirmed practice sources
Score-reference targets	3 configured
Reference training	3 source-confirmed score targets, 22 pitch/rhythm windows, title-labeled calibration sources for future scale/arpeggio uploads, and 4 official YouTube Data API public-reference queries kept separate from repertoire claims and audio evidence
Analyzed practice days	41 indexed days; 3 practice-time measured daily records; 3 audio-evidence records; 2 score-audio-only records; 1 detected-note-fragment record; 3 confirmed source labels, score targets, and a source-backed Wieniawski nine-note symbolic opening phrase now corrected to `A5 G5 F5 A5 G5 F5 A5 G#5 F5`
Machine pitch events	Detected-active-section onset-aware pYIN plus onset-bounded spectral pitch rescue for fast passages, audio-agreement counts, pitch-sanity filtering, repeated-pitch-collapse guarding, practice-coverage fields, title-labeled calibration handling, and fingerprints stored only when quality gates pass; broad candidate micro-transcription is withheld, while detected note series are preserved in the transcription PDF and promoted to note-match groups only after pitch-class sequence agreement with an available score-derived or reference-audio sequence
Specific observations	Rendered only when grounded in reliable score-linked evidence or repeated score-free technique-pattern evidence
Note-match windows	0 accepted score-linked phrases after the five-note 2026-05-03 Scherzo-Tarantelle phrase was rejected; D/A audio-only detections remain useful note evidence but are not score-location matches
Transcription solve tracker	Current code state after the five-note rejection lock: 0 accepted five-note source-score phrases; 0 accepted measure-level matches; 0 exact score-aligned windows; 0 score-coordinate heat-map fragments; 0 verified source snippets; hidden transition candidates and score-glyph review packets remain implementation support, not accepted evidence
Daily evidence surface	Local sample playback only for violin-positive windows, notation withheld until reliable note/rhythm verification, score snippets hidden until played-to-score agreement, and explicit whole-session transcription limit
Curtis-level completion	Not scored when the available evidence only identifies repertoire
Current focus	Capture a playable audition-view take
Current constraint	Record 30 seconds of sustained tone
Boundary	Curtis admission cannot be predicted from current samples

2.1 Media indexing

The source inventory stores platform, title, publication time, duration, view count, candidate reasons, media kind, and blocker state. Dated long-form practice logs are treated as high-value longitudinal records because the title contains a practice date and the duration preserves session-level evidence. The uploaded-video ledger starts at the `violin 1` marker and counts only violin-numbered or date-titled practice logs, so unrelated long public videos are retained in inventory without being silently counted as violin practice. This ledger is not the active-practice total. Active practice is measured only after media processing identifies windows where violin playing is actually occurring.

2.2 Review evidence

The review layer stores section-level findings with a dimension, evidence source, judgment, and practice constraint. The current dimensions include tone, intonation, shifts, time, articulation, and audition delivery. Findings can be strong, needing work, or unjudged. The unjudged state is part of the system design: if the sound is not available, the hand is obscured, the camera angle hides a transition, or still frames do not support a timing claim, the system should not convert weak evidence into a confident critique.

3 First readout

The first live readout is experimental and intentionally conservative. The strongest immediate value is that the system has already converted a large, timestamped practice archive into a queryable corpus with captured media samples, stored section reviews, and title-confirmed practice records. The home screen has three top-level surfaces: a paper card, analyzed practice days, and repertoire, followed by a compact transcription-completion tracker at the bottom. The first evidence row now reports total practice time first, then the video already checked for violin-playing, then the 234 h 57 min YouTube practice archive; this prevents the public practice ledger from being confused with either transcription coverage or uploaded-video duration. Each analyzed day groups same-day videos, preserves uploaded duration, and names confirmed or uncertain piece evidence without promoting guesses. Playable local media samples, paired audio/video evidence windows, and practice-time credit are now withheld unless violin playing is detected in the source window. The owner-sync sampler now reuses locally cached captured media, uploads cached violin-positive windows before blind new captures, resumes interrupted queue-only runs from cached pending-window media including non-playing windows, consumes active-scan pending windows before generic expansion or blind sampling, has a queue-only mode for chronological pending-window coverage, expands around already-found violin-positive windows, and can switch into full-check mode to walk every 90-second window in order instead of staying limited to sampled windows; the live analyzed-days surface currently exposes violin-playing practice-time evidence for 5/1, 5/2, and 5/3, and the first twenty-five `violin 1` windows are now represented as checked coverage, with three non-playing/unclear windows and twenty-two active windows. Broad sheet-music notation is also withheld until reliable note/rhythm verification, so audio evidence cannot appear as a failed transcription. The current visible note evidence includes exact audio-checked D4 and A4 fragments; the five-note Scherzo-Tarantelle phrase from 5/3 is rejected and stays candidate-only. Single-note Scherzo-Tarantelle score crops remain rejected. A multi-note score panel now renders only when the detected MIDI sequence, the parsed score MIDI sequence, the written spelling context, a non-generated source-score crop, and musician review agree on the same visible range; generated notation SVGs are no longer allowed to stand in for the score panel. A source-confirmed piece title creates a score target, and a detected pitch-class sequence match can show the played clip and transcription, but broad score-region estimates and visually unverified crops do not render as sheet-music snippets. The daily record includes a compact source-label field so a missing or wrong day title can be corrected without letting that correction masquerade as score alignment. The analyzed-days surface no longer promotes broad candidate transcription while candidate micro-transcription fails audible match review; those windows remain hidden support data or audio evidence only. Active scans with no confirmed piece are treated as possible repertoire or score-free technique exercise evidence rather than as failed score matching. The repertoire surface is evidence-backed: an entry appears only when a daily record carries confirmed source evidence, and each entry keeps practice dates, clips, score targets, practice time when measured, and current blocker limits. Generic etude/caprice evidence remains outside the piece-title field, and possible or uncorroborated repertoire labels remain checkable against the source window. These statements are not generic pedagogy; they are extracted from stored review findings, sample indexes, source corrections, public-domain score targets, pitch-quality checks, and the uploaded-video ledger.

After the 2026-05-15 10:05 PM EDT five-note rejection lock, the visible Scherzo-Tarantelle evidence surface has no accepted score-linked group. The previously shown `Bb4 D5 C5 Bb4 D5` group is blocked from accepted display. The acceptance rule is deliberately severe: if a source crop, note label, transcription glyph, audio-derived MIDI sequence, or musician review disagrees, the group must drop back to hidden candidate data instead of appearing as a match.

Table 2. Current evidence classes and limits.

Class	Stored signal	Limit
Audio	Tone, intonation, shifts, time, articulation	Some audio reviews fail or cannot inspect the excerpt cleanly
Video	Setup, posture, bow lane, visible hand/bow evidence	Still frames can hide motion, sound, and transition accuracy
Metadata	Dates, duration, channel, view count, practice-candidate labels	Metadata cannot judge playing quality by itself
Progress plan	One focus, practice constraint, short session plan	The plan is a next experiment, not an outcome prediction

Current transcription limit. Machine note output now has three display gates. First, the media window must be classified as violin-positive before it can appear as playable evidence or count toward practice time. Second, the note stream must audibly match the paired audio before any notation can render. Third, repertoire passages need score alignment, while score-free exercises need repeated-pattern grouping before they can become full practice transcription. Source-confirmed score targets are also display-gated: broad score snippets are hidden unless the corresponding played window has explicit score-location agreement, and single-pitch score crops require explicit visual note verification before they can render. Score-note sequences are now source-gated before matching: a scanned PDF plus guessed `scorePitchClassSequences` is ignored unless the note sequence has an explicit symbolic-score source, manual source confirmation, MusicXML/source-confirmed status, or exact score-location metadata. Curtis now includes a symbolic-score parser for inline MusicXML, local `.musicxml`, and local `.mxl` targets. That path extracts score notes, measures, pitch classes, written note names, octave-aware MIDI values, and duration fields. The parser now preserves written flats such as B-flat for score staff placement while matching their enharmonic pitch class to detected audio, so a source-score B-flat can no longer be rendered as an A-sharp staff position. The Scherzo-Tarantelle target now carries a source-backed nine-note symbolic opening phrase from the local IMSLP solo part, but the latest reviewed five-note crop for `Bb4 D5 C5 Bb4 D5` is rejected and blocked from accepted score display. A visible score match normally requires at least five contiguous played pitch classes and at least three distinct pitch classes; target-specific source excerpts may still use a four-note measure-level acceleration gate, but no current Scherzo-Tarantelle phrase counts as accepted short-phrase progress. One-note overlaps and repeated single-pitch runs are still rejected as score evidence. Hidden fast-transition candidates and reference-audio pitch traces now also require second-pass audio agreement for every matched note before they can create a candidate phrase; weak one- or two-note overlaps are withheld as search data, not progress. When the symbolic gate passes, generated notation can render candidate transcription support, but the score panel is accepted only if it uses an actual source-score crop whose visible noteheads, register, written spelling, exact range, paired audio, transcription, and musician review match; a generated SVG excerpt can no longer masquerade as the actual score. Source-verification targets now require exact detected MIDI values and per-note audio agreement before even the source-check status can become a score-sequence verdict; pitch-class overlap alone is recorded only as search context. All current Scherzo-Tarantelle single-pitch crops and the latest five-note 5/3 crop remain rejected and withheld because review found that the visible score evidence did not match the displayed transcription. These failures are now regression fixtures. Unverified room, setup, laptop, silence, or non-playing samples remain withheld, and unverified pitch traces remain hidden support data rather than visible notation. A spectral-onset rescue path estimates pitch inside short onset-bounded violin segments when the pYIN path collapses, while the repeated-note trace guard prevents fast or complex playing from becoming a dominant repeated pitch, records the quality mode, and keeps unusable traces out of accepted score evidence. The current micro-transcription path stores candidate short contiguous runs, but it does not render them as sheet music or count them as solved transcription until the displayed notes pass paired-audio listening review. The transcription-run PDF preserves all stored detected note series for the day. Source-score promotion uses exact MIDI sequence agreement plus paired-audio agreement; looser pitch-class grouping remains candidate-only and reference-audio matches alone are not treated as score notation. A separate audio-checked fragment gate can render a note when pYIN and YIN agree on the same semitone over a short stable window; the present live examples are D4 and A4, not a full passage and not a score-location match. Full-session transcription remains incomplete until every playing section in the long-form practice videos is extracted, verified as violin-positive, verified against the audio, and aligned either to a score section or to a repeated technique pattern.

The current rejection proves the strict lane is still not strong enough to solve long phrases. Curtis can now keep false score evidence out of the accepted surface, but it still needs a newly verified phrase that carries many consecutive notes through source-score parsing, actual-source crop selection, local audio/video pairing, transcription rendering, musician review, and regression tests without showing a wrong-note score crop. The next unsolved step is length: the same gate must work for many consecutive notes, then measures, repeated attempts, rhythm, and full active-playing windows.

4 Discussion

Curtis Media Review is intentionally conservative. It does not rank the player against applicants, infer admissions probability, or convert every video into a claim. Instead, it keeps a running technical record. The sampler now treats violin presence as a hard evidence boundary: playable clips and practice-time credit appear only after violin playing is detected in a media window, while notation waits for reliable note/rhythm verification and either score alignment or score-free pattern grouping. This makes the system useful even before it becomes musically sophisticated: it can preserve practice volume, surface recurring constraints, identify missing evidence, and make subsequent recordings easier to compare against earlier ones.

The main open problem is repertoire identification, score-free exercise representation, and evidence quality. YouTube metadata can identify candidate videos and public labeled reference sources, and the current owner-sync path can provide direct audio/video samples, but unclear camera framing, missing score/title context, improvised technical patterns, and short excerpts still prevent reliable piece naming. The backend therefore keeps generic labels out of the piece-title field, stores them as candidate evidence, requires source-window evidence for confirmed titles, and marks weak evidence explicitly. The current training ledger separates source-confirmed labels from title-labeled calibration uploads, public YouTube reference metadata, independent audio-title matches, score-free or piece-pending active windows, failed unverified machine pitch/rhythm windows, failed pitch-collapse windows, withheld candidate micro-transcription windows, audio-checked single-note fragments, detected note series, pitch-class reference matches, and true score-aligned windows. It also stores reference targets for the three confirmed source labels across two works so later passes can ask for section-level matching instead of repeating broad title guesses. The sampler classifies candidate windows using active audio, violin-range voiced pitch, onset density, pitch-class diversity, spectral features, and minimum practice-duration gates before a sample can become visible evidence. The transcription path converts sampled media to mono audio, finds audio-active playing sections, runs pitch tracking only on those sections, normalizes and harmonic-isolates each section for pitch tracking, estimates monophonic violin pitch with an onset-aware pYIN pitch track bounded to the violin range, detects onset boundaries so repeated attacks on the same pitch are not silently merged into one sustained note, runs a second spectral pitch estimate inside short onset-bounded segments when fast playing defeats pYIN, compares pitch diversity across candidate note streams, checks whether selected note events agree with an independent spectral or pYIN/onset detector, rejects out-of-range pitch artifacts, requires sustained pitch changes for pitch-track segmentation, removes very short low-confidence pitch blips, marks isolated octave-flip corrections as uncertain, rejects repeated-pitch collapse, estimates tempo, stores detected note series for audit/PDF review, and builds pitch-class, interval, rhythm, and contour fingerprints only when quality gates pass. Daily records now carry transcription quality, material status, and practice-coverage states, and the primary interface withholds media playback unless the source window is violin-positive, withholds broad machine notation until note/rhythm verification, displays only audio-checked fragments when the stricter gate passes, hides loose reference-audio matches from the daily row, shows a source-score crop only when the highlighted score note has explicit visual note verification, and withholds broader score snippets until exact score-location alignment exists. A stronger system should align repertoire events against a symbolic or optical score representation, group score-free technical patterns by repeated note/rhythm fingerprints, identify movement sections or measure ranges when a score exists, compare against reference recordings available through permitted media, and then judge articulation, timing, tone, and tempo against the aligned passage or repeated pattern.

The broader experiment is whether a practice system can make musical development legible over time. A useful Curtis system should show whether tone, intonation, rhythm, articulation, shifts, memory, endurance, and audition delivery are improving across dated recordings. It should also preserve the source evidence behind each claim so that the dashboard does not become motivational text detached from the actual playing.

5 Roadmap to solved long-phrase transcription

This section is the working plan for making Curtis able to produce long, trustworthy audio–video–transcription–score evidence from Alan's actual violin practice. It is intentionally part of the paper rather than a separate planning note. Whenever Curtis makes real progress toward full-session active-practice measurement, accurate transcription, score matching, heat maps, repertoire updates, or Curtis-level observation, this PDF should be updated with the new state, the acceptance evidence, and the remaining blocker.

5.1 Current tracker state

The tracker was refreshed through the 2026-05-16 source-position and exact-MIDI source-target update. The system has a real media ledger, a deployed active-practice scan route, a working owner-sync path for captured media, a visible 100-point transcription-completion tracker with precise weighted progress, stored user-correction benchmarks, a persistent truth workbench for accepted/rejected/pending audio-score-transcription items, a local source-backed Wieniawski score PDF, a nine-note source-backed Wieniawski symbolic score opening phrase, a source-verification target queue, source-target score triage, an unaccepted score-glyph verification queue, an unaccepted score-note hypothesis queue, staff-level source review packets, a stricter evidence gate, and a hidden YIN fast-transition candidate trace for score matching. The Wieniawski opening source map is now `A5 G5 F5 A5 G5 F5 A5 G#5 F5`; the previous `D6 C6 Bb5 D6 C6 Bb5 D6 C#6 B5` and lower-octave `D5 C5 Bb4 D5 C5 Bb4 D5` maps are rejected and covered by regression tests. The 2026-05-03 Scherzo-Tarantelle five-note phrase `A#4 D5 C5 A#4 D5`, displayed as `Bb4 D5 C5 Bb4 D5`, is explicitly rejected after Alan's review and cannot display as accepted score evidence. The old A4/A5 single-note anchors remain rejected, and a one-note pitch anchor still cannot render unless the score note and octave are explicitly verified. The current top longer source target `A D D F# D` remains unaccepted because source-target checks now require exact detected MIDI values, per-note audio agreement, and a contiguous exact-MIDI run in the parsed MusicXML before a score phrase can display. The transition trace is not displayed as notation and now fails closed unless every matched fast-note candidate has a second detector agreement path, including spectral/onset agreement, sufficient confidence, and sufficient duration. Loose one- or two-note reference-audio coincidences no longer count as phrase progress, and candidate pitch-sequence groups no longer enter the daily `matchGroups` display bucket unless the exact source-score crop, visible note sequence, box centers, paired audio, transcription, score spelling, and accepted truth-evidence flags all pass. This is not solved long-phrase transcription yet; the current accepted score-linked phrase count is zero until a phrase survives audio, transcription, source-score, crop, and musician review.

Table 3. Progress tracker baseline for the long-phrase transcription goal.

Tracker item	Baseline state
Goal	Produce compact daily evidence groups: original score snippet, generated transcription, local audio, local video, heat-map context, and specific observation.
Current accepted transcription	0 score-linked transcription phrases are accepted after the five-note Scherzo-Tarantelle phrase was rejected. Audio-checked D/A fragments remain useful note evidence, but they are not score-location matches or long-phrase transcription.
Current accepted score match	0 actual-source score crops are accepted. The 2026-05-03 five-note Scherzo-Tarantelle crop, the old A4/A5 single-note anchors, and the previous D–B-flat–G–D score extraction are all rejected fixtures. The current `A D D F# D` longer source target is still rejected for now because it does not have an audio-agreed exact-MIDI run in the verified source MusicXML; 951 local score glyph candidates, 214 staff-position notehead hypotheses, 8 staff review packets, and 6 source-verification targets are queued for truth review but are not accepted score evidence.
Current archive coverage	234 h 57 min uploaded practice video across 48 strict-ledger videos and 41 practice days; 2 h 28 min checked; 2 h 21 min measured active violin; full chronological coverage still pending.
Current estimate	223 h 5 min total active practice estimated from checked windows; provisional and sample-biased until the full chronological scan covers the archive.
Current user burden to remove	Alan is still forced to inspect whether a note is real, whether a score box points to the correct pitch, whether audio matches notation, and whether practice time means playing rather than uploaded duration.
First tracker update	This PDF now defines the full implementation plan and becomes the required progress record for future Curtis work.

Progress log

2026-05-13 11:57 AM EDT code update Backend coverage ledger, evidence-correction route, and benchmark-fixture scaffold added. Checked video time now includes sampled non-playing windows; total practice time remains measured active playing only. Live deployed coverage now reports 213 h 23 min uploaded video, 2 h checked video, 53 min 14 s measured active practice, and 211 h 23 min unchecked. Wrong score-note corrections can be stored as rejected benchmark cases. Full archive scan and long-phrase transcription remain unsolved.

2026-05-13 12:33 PM EDT code update Low-cost active-practice scan route implemented. The new scan persists active violin-playing sub-intervals from stored violin-positive media samples, keeps checked non-playing samples separate from practice time, exposes `/api/curtis/active-practice-scan`, adds an active-scan run endpoint, queues remaining strict-ledger windows for owner media capture, and feeds persisted active intervals into daily records and the coverage ledger. Local unit tests verify active sub-interval extraction, negative-sample withholding, and coverage accounting.

2026-05-13 12:59 PM EDT deployed scan update The active-scan endpoint was deployed and run against the live Curtis state. It persisted 214 active sub-intervals from 80 stored media-sample results, representing 41 min 42 s of scan-derived active audio, and queued 250 strict-ledger windows for later owner-media capture. The coverage ledger reports 1 h 57 min 24 s measured active practice from 2 h checked and a 208 h 46 min provisional archive estimate. A regression fix now caps active practice by checked source coverage and makes active-scan results supersede noisy transcription-duration evidence where scan results exist. The full 213 h 23 min archive scan and long-phrase transcription remain unsolved.

2026-05-13 03:26 PM EDT queue update The owner-media sync now consumes the active-scan pending-window queue before choosing expansion or blind sample windows. The active-scan status endpoint can expose a bounded pending-window preview, and the run endpoint returns a compact scan/coverage payload rather than the full operations document. Regression tests now verify that queued active-practice windows are selected first and that pending-window previews preserve total pending counts while limiting returned rows. This does not solve long-phrase transcription; it closes the next M1 plumbing gap so the chronological archive scan can keep advancing without Alan manually choosing windows.

2026-05-13 03:37 PM EDT live owner-sync update A one-window owner-sync pass uploaded one cached violin-positive 5/3 sample, `Njh8_zq9_DM-10725` from `*10725-10815`. The follow-up active scan selected that new sample and recorded one new active interval. Live coverage is now 81 active-scan sample results, 215 active sub-intervals, 42 min 9 s scan-derived active audio, 2 h 1 min 30 s checked video, 1 h 58 min 54 s measured active practice, and a 208 h 49 min provisional estimate. The first chronological pending queue item remains `violin 1` at `*0-90`; this run advanced cached-media coverage, not the chronological queue.

2026-05-13 04:05 PM EDT completion tracker and queue-only update Curtis now exposes a bottom-page transcription completion tracker backed by a 100-point implementation gate checklist. The tracker reports archive coverage, accepted long phrases, exact score windows, scan queue state, completed gates, remaining gates, and the next concrete action. A queue-only owner-sync mode was added so the active-scan pending-window list can be consumed without falling back to cached positives, expansion windows, or blind sampling. A queue-only run uploaded `otHfHMgDo2g-0` from `violin 1` at `*0-90`; the sample was classified as `not_violin_or_unclear` and therefore did not add practice time. The follow-up active scan reports 82 sample results, 72 active-violin samples, 10 checked non-violin samples, 215 active sub-intervals, 2 h 1 min 30 s checked video, 1 h 58 min 54 s measured active practice, and a 208 h 49 min provisional estimate. The next chronological pending window is `violin 1` at `*90-180`.

2026-05-13 05:44 PM EDT queue resume and practice-time ceiling update The next queue-only owner-sync attempt captured `violin 1` windows `*90-540`. The wrapper timed out after the uploads, revealing that cached pending-window media was excluded from queue-only resume. The owner-sync helper now resumes cached queue windows, including cached non-playing windows, and tests cover that path. The follow-up active scan reports 87 sample results, 77 active-violin samples, 10 checked non-violin samples, 238 active sub-intervals, 2 h 3 min checked video, 1 h 55 min 30 s measured active practice, and a 200 h 23 min provisional estimate. Practice time is now hard-capped by checked source coverage, so unchecked transcription remnants cannot make active practice exceed checked video. The next chronological pending window is `violin 1` at `*540-630`.

2026-05-13 06:01 PM EDT chronological queue update A one-window queue-only owner-sync pass uploaded `otHfHMgDo2g-540` from `violin 1` at `*540-630`. The sample was classified as `violin_positive` with a sampler score of 68.6. The follow-up active scan selected that sample, recorded 4 new active sub-intervals, and reports 88 sample results, 78 active-violin samples, 10 checked non-violin samples, 242 active sub-intervals, 2 h 4 min 30 s checked video, 1 h 57 min measured active practice, and a 200 h 32 min provisional estimate. The live accounting check passes because active practice remains below checked video. The next chronological pending window is `violin 1` at `*630-720`.

2026-05-13 06:26 PM EDT chronological queue update A one-window queue-only owner-sync pass uploaded `otHfHMgDo2g-630` from `violin 1` at `*630-720`. The sample was classified as `violin_positive` with a sampler score of 60.3. The follow-up active scan selected that sample, recorded 5 new active sub-intervals, and reports 89 sample results, 79 active-violin samples, 10 checked non-violin samples, 247 active sub-intervals, 2 h 6 min checked video, 1 h 58 min 30 s measured active practice, and a 200 h 41 min provisional estimate. The live accounting check passes because active practice remains below checked video. The next chronological pending window is `violin 1` at `*720-810`.

2026-05-13 07:55 PM EDT chronological queue update A one-window queue-only owner-sync pass uploaded `otHfHMgDo2g-720` from `violin 1` at `*720-810`. The sample was classified as `violin_positive` with a sampler score of 55.2. The follow-up active scan selected that sample, recorded 3 new active sub-intervals, and reports 90 sample results, 80 active-violin samples, 10 checked non-violin samples, 250 active sub-intervals, 2 h 6 min checked video, 2 h measured active practice, and a 203 h 14 min provisional estimate. The live accounting check passes because active practice remains below checked video. The next chronological pending window is `violin 1` at `*810-900`.

2026-05-13 09:25 PM EDT scan and benchmark update Queue-only owner sync consumed the next two chronological `violin 1` windows, `*810-900` and `*900-990`. Both were classified as `violin_positive`. The follow-up active scan reports 92 sample results, 82 active-violin samples, 10 checked non-violin samples, 257 active sub-intervals, 2 h 10 min checked video, 2 h 3 min measured active practice, and a 201 h 7 min provisional archive estimate. The next chronological pending window is `violin 1` at `*990-1080`. The same pass stored three real user-reviewed score-match failures as benchmark corrections: `A4` displayed against a B-position score crop, `A4` displayed against a `G4` crop, and `A5` displayed against a `G5` crop. These benchmark fixtures moved the implementation tracker from 29 percent to 31 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-13 10:05 PM EDT scan and exact-percent update Queue-only owner sync consumed `violin 1` windows `*990-1080` and `*1080-1170`. Both were classified as `violin_positive`. The follow-up active scan reports 94 sample results, 84 active-violin samples, 10 checked non-violin samples, 269 active sub-intervals, 2 h 13 min checked video, 2 h 6 min measured active practice, and a 201 h 24 min provisional archive estimate. The next chronological pending window is `violin 1` at `*1170-1260`. The rounded implementation label remains 31 percent, but exact weighted progress moved to 31.13 percent and is now displayed so small coverage gains are visible. Exact score windows and accepted long phrases remain at 0.

2026-05-13 11:14 PM EDT scan and precision update Queue-only owner sync consumed `violin 1` windows `*1170-1260` and `*1260-1350`. The first was classified as `violin_positive`; the second was classified as `not_violin_or_unclear`, which adds checked non-playing coverage without adding practice time. The follow-up active scan reports 96 sample results, 85 active-violin samples, 11 checked non-violin samples, 283 active sub-intervals, 2 h 16 min checked video, 2 h 7 min measured active practice, and a 199 h 19 min provisional archive estimate. The next chronological pending window is `violin 1` at `*1350-1440`. The implementation tracker now computes exact weighted progress from unrounded gate points, so the live label should read 31.128 percent after deployment. Exact score windows and accepted long phrases remain at 0.

2026-05-13 11:38 PM EDT scan update Queue-only owner sync consumed `violin 1` windows `*1350-1440` and `*1440-1530`. Both were classified as `violin_positive`. The follow-up active scan selected both samples, recorded 11 new active sub-intervals, and reports 98 sample results, 87 active-violin samples, 11 checked non-violin samples, 294 active sub-intervals, 2 h 19 min checked video, 2 h 10 min measured active practice, and a 199 h 37 min provisional archive estimate. The next chronological pending window is `violin 1` at `*1530-1620`. Exact weighted implementation progress is now 31.131 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-13 11:55 PM EDT scan update Queue-only owner sync consumed `violin 1` windows `*1530-1620` and `*1620-1710`. The first was classified as `not_violin_or_unclear`; the second was classified as `violin_positive`. The follow-up active scan selected both samples, recorded 6 new active sub-intervals, and reports 100 sample results, 88 active-violin samples, 12 checked non-violin samples, 300 active sub-intervals, 2 h 22 min checked video, 2 h 12 min measured active practice, and a 197 h 40 min provisional archive estimate. The next chronological pending window is `violin 1` at `*1710-1800`. Exact weighted implementation progress is now 31.134 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-14 12:09 AM EDT scan update Queue-only owner sync consumed `violin 1` windows `*1710-1800` and `*1800-1890`. Both were classified as `violin_positive`. The follow-up active scan selected both samples, recorded 13 new active sub-intervals, and reports 102 sample results, 90 active-violin samples, 12 checked non-violin samples, 313 active sub-intervals, 2 h 25 min checked video, 2 h 15 min measured active practice, and a 197 h 59 min provisional archive estimate. The next chronological pending window is `violin 1` at `*1890-1980`. Exact weighted implementation progress is now 31.136 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-14 12:35 AM EDT scan update Queue-only owner sync consumed `violin 1` windows `*1890-1980` and `*1980-2070`. Both were classified as `violin_positive`. The follow-up active scan selected both samples, recorded 9 new active sub-intervals, and reports 104 sample results, 92 active-violin samples, 12 checked non-violin samples, 322 active sub-intervals, 2 h 28 min checked video, 2 h 18 min measured active practice, and a 198 h 18 min provisional archive estimate. The next chronological pending window is `violin 1` at `*2070-2160`. Exact weighted implementation progress is now 31.139 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-14 03:00 AM EDT scan update Queue-only owner sync consumed `violin 1` windows `*2070-2160` and `*2160-2250`. Both were classified as `violin_positive`. The follow-up active scan folded both samples into the checked ledger and reports 106 sample results, 94 active-violin samples, 12 checked non-violin samples, 331 active sub-intervals, 2 h 31 min checked video, 2 h 21 min measured active practice, and a 198 h 36 min provisional archive estimate. The next chronological pending window is `violin 1` at `*2250-2340`. Exact weighted implementation progress is now 31.142 percent. Exact score windows and accepted long phrases remain at 0.

2026-05-14 03:23 AM EDT one-measure accelerator update Curtis no longer hardcodes accepted long phrases to zero. The tracker now derives accepted long-phrase count from daily match groups that pass all of the following gates: score location verified, exact score-location status, at least a five-note phrase run, local playable media, and not merely a single-pitch anchor. Regression tests now prove that a verified one-measure symbolic-score phrase with media increments the long-phrase count, while single-note anchors, unverified reference matches, and matches without media do not. This is implementation progress toward a faster path, not solved musical evidence yet: the live percent remains 31.142 percent until Curtis has a verified symbolic score measure or more scanned archive coverage. The fastest route to visible musical progress is now to verify one Scherzo-Tarantelle symbolic score measure and run the existing phrase matcher across the hidden detected note series already stored for 2026-05-03.

2026-05-14 04:32 AM EDT source-score acceleration update Curtis now carries the Wieniawski Scherzo-Tarantelle solo score PDF as a local source asset rather than depending on weak scanned crops or a network fetch during matching. The score-asset layer reports whether a local source PDF is present, the daily score-reference audit counts local source PDFs separately from symbolic notes, and the 100-point tracker awards source-truth progress for a local source score without counting it as an accepted phrase. Regression tests verify that this raises the score-source readiness state while keeping accepted long phrases, exact score windows, and symbolic score notes at 0. This is the necessary acceleration layer for long-phrase work: the next step is no longer finding a score image, but converting one local source-score measure into verified symbolic notes and running the existing phrase matcher over stored detected note series.

2026-05-14 05:23 AM EDT live symbolic-measure update The deployed Curtis API reported 43.642 percent exact progress, 1 provisional measure match, 1 provisional score-aligned window, 0 accepted long phrases, 1 local score source, and 8 symbolic score notes counted across the two

Wieniawski practice-day targets. This state is now superseded: the later source-truth correction rejected the provisional `D A# G D` source-score claim because the local IMSLP solo part did not support that score-note sequence. The useful implementation gain from this pass was the source-symbolic note gate and score/audio/video/transcription lane, not the specific provisional score phrase.

2026-05-14 05:40 AM EDT source-target accounting update The deployed Curtis API reported 45.142 percent exact progress after the tracker began counting configured source score targets from the reference-training state instead of only local PDF files. That accounting layer remains useful, but the musical evidence count from this pre-correction state is superseded by the 07:52 AM source-truth correction. This makes the progress meter more honest and removes a hidden undercount, but it does not solve phrase transcription; the next implementation step is extending corrected accepted symbolic measure gates into longer matched note runs.

2026-05-14 06:52 AM EDT phrase-candidate ranking update The deployed Curtis API reported 47.642 percent exact progress after the matcher stopped returning the first scan-order matches and instead ranked all detected-series matches by exact score status, score-derived status, distinct pitch-class content, and run length. That state exposed 12 pitch-sequence groups, 1 phrase candidate, 1 provisional measure match, 1 provisional score-aligned window, and 0 accepted long phrases. The highest-ranked pending phrase candidate was `D D D# D D A# G`. This candidate was not accepted score evidence; it was a focused source-score verification target. The later strict evidence gate supersedes this provisional count.

2026-05-14 07:08 AM EDT tracker visibility update Curtis now returns the highest-ranked pending phrase target as structured tracker state and displays it in the implementation card detail as `pending: D D D# D D A# G`. The visible interface therefore shows not only that one phrase candidate exists, but also which detected pitch-class sequence should be source-score verified next. This does not add accepted musical evidence or change the 47.642 percent full-solve score; it removes a product friction point by making the next verification target visible without presenting it as a solved score match.

2026-05-14 07:52 AM EDT source-truth correction update The Wieniawski symbolic source opening was corrected against the local IMSLP solo-violin score. The old accepted `D A# G D` match was demoted because the source phrase is `D C Bb D C Bb`; it can remain only as an unverified reference-sequence candidate, not accepted score evidence. The deployed provisional groups matched detected pitch classes `A# D C A#` to source score pitch classes `A# D C A#`, displayed a source-score crop for that phrase, and retained generated notation only as fallback. The tracker then reported 48.642 percent, 2 provisional measure matches, 2 provisional score-aligned windows, 12 pitch-sequence groups, 1 pending phrase candidate, 3 score targets, 1 local score PDF, 6 unique symbolic score notes, 1 source snippet candidate, and 0 accepted long phrases. The later strict evidence gate supersedes these provisional score counts.

2026-05-14 08:29 AM EDT first source-backed phrase update The Scherzo-Tarantelle source excerpt was extended by one verified score note, from `D C Bb D C Bb` to `D C Bb D C Bb D`, using a widened crop from the local IMSLP solo-violin PDF rather than the annotated teaching image. That seven-note source phrase let the stored 2026-05-03 detected series match the contiguous five-note run `A# D C A# D` against source score notes `Bb4 D5 C5 Bb4 D5`. The deployed API then reported 55.642 percent exact progress, 1 provisional long-phrase candidate, 2 provisional measure matches, 2 provisional score-aligned windows, 12 pitch-sequence groups, 7 unique symbolic score notes, and 1 source snippet candidate. The useful gain was the source-backed path; the later visual review rejected the crop as accepted score evidence, so the strict current state is again 0 accepted long phrases.

2026-05-14 01:50 PM EDT score-coordinate heat-map update Provisional symbolic score phrase matches were connected to score-coordinate heat-map fragments. The 2026-05-03 Scherzo-Tarantelle phrase `A# D C A# D` was mapped to `mm. 2-4` with source score notes `Bb4 D5 C5 Bb4 D5`, local video, local audio, and an IMSLP score crop. The deployed API reported 58.136 percent exact progress, 1 provisional long-phrase candidate, 2 provisional measure matches, 2 provisional score-aligned windows, 2 provisional score-coordinate heat-map fragments, 12 pitch-sequence groups, 7 unique symbolic score notes, and 1 source snippet candidate. This was a heat-map and evidence-organization advance; the later strict evidence gate blocks those fragments from counting as accepted evidence.

2026-05-14 02:15 PM EDT source-verification target update Curtis now separates the next longer note-run target from accepted score evidence. The tracker exposes one source-verification target, `D D D# D D A# G`, only because it is a seven-note run from local playable practice media with enough pitch diversity to be worth checking against the score. It remains reference-audio evidence, not accepted score evidence. The source target appears in the tracker and API so the next pass can verify exact local

IMSLP score notes and coordinates instead of slowly searching from scratch or accidentally promoting a wrong crop. The deployed tracker state then included provisional score artifacts, but the current strict evidence gate counts 0 accepted long phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 score-coordinate heat-map fragments, and 0 verified source snippets until source review passes.

2026-05-14 02:51 PM EDT source-target triage update Curtis now checks four-note-or-longer source targets against the verified symbolic score map before promotion. The current top target `D C G A` was compared against the verified source excerpt `D C A# D C A# D`. The best contiguous overlap is two notes, so the target is recorded as `source_score_sequence_not_found` and remains reference-audio evidence only. The tracker then reported 58.633 percent exact progress, 3 source-target score checks, 0 verified source targets, and the next action: extend the verified IMSLP symbolic score map before promoting this sequence. This improves the long-phrase pipeline by making measure-sized candidates fail fast and explicitly instead of appearing as misleading score matches.

2026-05-14 03:49 PM EDT score-map workbench update Curtis now has an explicit source-map acceleration layer for the local IMSLP Wieniawski solo part. A deterministic extraction tool renders page 2 of the local score PDF, detects staff-line groups, removes staff lines, extracts connected score glyph candidates, and writes a JSON verification queue. The current queue contains 779 unaccepted score glyph candidates across 7 staves. These candidates are not pitch labels, not accepted score notes, and not displayable score evidence. They exist to make the next work cycle faster: verify candidate glyphs against the local PDF, convert only verified notes into MusicXML, then run source-target matching against verified symbolic notes. The tracker now reports 59.633 percent exact progress after awarding score-truth credit for the verification queue while preserving 0 verified source targets for the current `D C G A` source target.

2026-05-14 04:24 PM EDT note-hypothesis acceleration update The source-map workbench now filters the local IMSLP glyph queue into likely notehead candidates and assigns staff-position pitch hypotheses using the G-minor key signature. The current queue contains 172 unaccepted score-note hypotheses across 7 staves in addition to the 779 raw glyph candidates. These hypotheses are explicitly not accepted score evidence: they use staff position only, may include false positives, and cannot create a score snippet, pitch label, or match card until source-reviewed into MusicXML. The implementation value is speed. The next pass can review a note-labeled queue instead of starting from anonymous glyph boxes, which should make extending the Scherzo-Tarantelle symbolic map to full measures and longer phrases much faster. The tracker now reports 61.133 percent exact progress after awarding limited score-truth credit for the note-hypothesis queue while preserving 0 verified source targets for the current `D C G A` source target.

2026-05-14 05:31 PM EDT staff-review packet update The score-map detector was changed from a single-threshold staff finder to a multi-threshold merged staff finder after review showed that the previous queue skipped a source staff. Page-2 coverage now reports 951 unaccepted score glyph candidates and 214 unaccepted notehead hypotheses across 8 staves. The same pass writes 8 staff-level review-packet images under the local score assets folder, with each packet showing the scanned IMSLP staff plus numbered hypothesis boxes. The packets are not accepted score evidence and do not change any match card by themselves; they exist to make the next MusicXML pass faster and less error-prone by turning source review into a bounded staff-by-staff checklist. The deployed tracker now reports 64.135 percent exact progress after awarding score-truth credit for complete source-review packet generation while preserving 0 verified source targets for the current `D D D# D D A# G` source target, whose verified symbolic-score overlap is still only 1/7.

2026-05-14 06:15 PM EDT score/transcription visual-agreement update The public 2026-05-03 match had a real failure: the score panel showed a broad opening source crop even though the claimed transcription phrase was the narrower five-note slice `Bb4 D5 C5 Bb4 D5`. Curtis then added a separate score-display gate: a score panel could render an actual source crop only when the crop's reference start and end exactly matched the phrase, and broad covering crops were withheld. The symbolic matcher also began requiring MIDI-level equality between detected notes and parsed score notes, not only pitch-class similarity, before it could create a score-visual lock. The deployed tracker reported 64.878 percent exact progress and provisional score locks, but later visual review showed the crop still was not trustworthy enough; the current strict gate counts this phrase as pending, not accepted.

2026-05-14 07:04 PM EDT actual-source score-gate update The previous visual-agreement fix still left one unacceptable loophole: a generated notation SVG could appear in the accepted score panel even though Alan asked for the actual piece. Curtis therefore required a score panel to use a non-generated source-score crop for the same reference range. The five-note 2026-05-03 Scherzo-Tarantelle match gained an exact-range source crop for `Bb4 D5 C5 Bb4 D5`; generated notation remained only as the transcription rendering. That deployment expected provisional actual-source score progress, but the later musician review rejected the crop and the current strict evidence gate blocks it from creating an accepted measure, long-phrase count, exact score-aligned window, score-coordinate heat-map fragment, actual-source score lock, or verified source snippet.

2026-05-14 09:45 PM EDT score-spelling visual-context update The latest reported mismatch exposed another gate that had been too weak: a transcription could be score-spelled as `Bb4 D5 C5 Bb4 D5` in data while the rendered notation panel omitted the source key signature, making the first and fourth notes look like unqualified natural-position B notes to a reader. Curtis now carries the source key signature into score-matched transcription rendering and requires both `scoreVisualRangeAgreement` and `scoreSpellingAgreement` before an actual-source score crop can count as accepted score evidence. The Scherzo-Tarantelle match remains a five-note fragment, but the visible transcription and visible score must now agree in musical spelling context rather than only MIDI or pitch class. This is a stricter acceptance gate, not a new long-phrase solve.

2026-05-15 12:29 PM EDT exact-note truth-gate update The reported failure mode was not only visual range: a score match could still be treated as acceptable if the pitch class agreed while the written note/register did not. The truth workbench now stores MIDI sequences for detected, accepted, and score notes; accepted score-truth items reject audio-score mismatches at exact note-and-octave level; and score-truth items cannot be accepted with score notes that omit octave/register. Single-note score anchors also require an explicit exact-note verification flag before the UI can render them. Regression tests now cover accepted A4–D5 truth, A4-versus-A5 rejection, score truth without octave rejection, exact pitch-anchor display, and visual-only pitch-anchor rejection. This does not create a new accepted phrase; it prevents Curtis from ever calling a pitch-class-only or visually guessed crop solved evidence.

2026-05-14 10:52 PM EDT visual-crop demotion update Alan's follow-up review showed that the notes visible in the 2026-05-03 source crop still did not match the transcription. Curtis now rejects the Scherzo-Tarantelle source crops as review material unless the visible noteheads, register, written spelling, and exact range are independently verified against the transcription. The five-note 5/3 phrase remains a symbolic phrase candidate, but it no longer creates an accepted score panel, accepted measure match, accepted long phrase, exact score-aligned window, score-coordinate heat-map fragment, score-visual lock, actual-source score lock, or verified source snippet. This is a regression lock: wrong score evidence is removed rather than cosmetically relabeled.

2026-05-15 12:45 AM EDT truth-workbench update Curtis now has a persistent truth-workbench API and compact implementation-status surface for accepted, pending, and rejected audio-score-transcription evidence. A truth item can store the source clip, practice day, piece, detected notes, accepted notes, score notes, score source, score location, score image, and gate state. Accepted truth items are rejected at write time if the accepted audio note sequence and score note sequence disagree. The backend and frontend score gates also now require explicit visible-note-sequence verification before a score panel can count as a score-visual lock, actual-source score lock, accepted measure match, or accepted long phrase. This prevents the exact failure Alan found: a visually wrong score crop cannot count just because the metadata says the pitch sequence should match. It still has 0 accepted long phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 score-coordinate heat-map fragments, 0 score-visual locks, 0 actual-source score locks, and 0 accepted truth clips.

2026-05-15 03:20 PM EDT exact-source phrase attempt Curtis temporarily promoted one actual-source Scherzo-Tarantelle phrase crop, `Bb4 D5 C5 Bb4 D5` from the local IMSLP page-2 solo-violin scan, aligned to the symbolic score range `mm. 2-4`. The attempt boxed five matched noteheads and stored source page pixel coordinates and exact visible note-plus-octave sequence. This pass was useful because it proved the crop extraction path could render actual score material, but it was later rejected by visual review because the score notes and boxes were not trustworthy enough to count as accepted evidence.

2026-05-15 03:40 PM EDT superseded deployment verification The deployed Curtis site served cache version `20260515-exact-source-phrase` on deployment `4fb78ce1`. That deployment still showed one score group for 2026-05-03, with detected notes `A#4 D5 C5 A#4 D5`, score-spelled transcription `Bb4 D5 C5 Bb4 D5`, source score notes `Bb4 D5 C5 Bb4 D5`, the exact-source crop asset, local video sample `Njh8_zq9_DM-9465`, and local audio from 6.618–7.674 s inside that sample. This state is now superseded by the 05:00 PM EDT strict evidence-gate pass because the displayed score crop did not survive musician visual review. The live practice-time surface reported 234 h 57 min uploaded archive video, 2 h 28 min checked video, 2 h 21 min measured active violin, 223 h 5 min estimated total active practice, and 232 h 28 min unchecked video. The estimate remains provisional until the full chronological active-practice scan covers every archive window.

2026-05-15 05:00 PM EDT strict evidence-gate update Curtis now rejects the 2026-05-03 Scherzo-Tarantelle source crop as accepted evidence unless the full truth chain passes: source note boxes centered on the visible noteheads, visible score notes matching the claimed written notes and octaves, transcription notes matching the paired audio, transcription notes matching the score notes, and an accepted truth-workbench item tying the audio, transcription, and score together. Backend, frontend, tracker, and tests now require those fields before a score snippet, accepted measure match, exact score-aligned window, actual-source score lock, or long-phrase count can appear. The deployed tracker reports 43.126 percent exact weighted progress, 0 accepted long phrases, 0 accepted measure matches, 0 exact score-aligned windows, 0 actual-source score locks, and 0 verified source snippets until a crop survives this stricter review.

2026-05-15 04:01 PM EDT practice-time API unification The daily-record API now consumes the same active-practice coverage ledger used by `/api/curtis/active-practice-coverage` before publishing top-level and per-day practice totals. This removes the stale split where one public route could show older transcription-derived totals while the coverage ledger showed 2 h 28 min checked video, 2 h 21 min measured active violin, 223 h 5 min estimated total active practice, and 232 h 28 min unchecked video. A regression test now verifies that daily records inherit canonical checked, active, estimate, unmeasured, and per-day scan totals from the coverage ledger.

2026-05-15 06:34 PM EDT strict-source correction attempt Curtis temporarily promoted a corrected actual-source crop for the same 2026-05-03 Scherzo-Tarantelle phrase. The detected sequence was `A#4 D5 C5 A#4 D5`; the displayed spelling was `Bb4 D5 C5 Bb4 D5`; the score-side sequence was the same octave-aware note sequence; and the score panel used a non-generated crop from the local Scherzo-Tarantelle solo part. This attempt is now superseded by Alan's later review: the phrase is rejected and cannot count as an accepted measure-level match, exact score-aligned window, score-coordinate heat-map fragment, score-visual lock, actual-source score lock, verified source snippet, or accepted five-note phrase.

2026-05-15 08:37 PM EDT fast-transition candidate update The next blocker was not another UI issue: the broader pYIN/onset stream in the 2026-05-03 Scherzo-Tarantelle window still collapses large parts of fast playing into repeated D/B-flat/G material, so a longer source-score phrase cannot be honestly promoted. Curtis now adds a hidden YIN frame-transition trace to every transcription run. This trace follows fast frame-level pitch changes and stores them as `scoreMatchCandidateNotes`; it is never rendered as accepted notation by itself. The daily matcher can search those hidden candidates against parsed source-score notes, but accepted score evidence still requires exact MIDI equality, exact written score spelling, visible source-score crop agreement, local audio/video, and no uncertain octave-corrected detected notes. Regression tests now prove that hidden transition candidates can feed the strict score matcher while uncertain octave-corrected candidates are rejected. A local check on the 5/3 `Njh8_zq9_DM-9465` clip produced 174 hidden fast-transition candidate notes with 8 pitch classes. After the five-note rejection, those candidates remain search data only and do not create accepted evidence. This is implementation progress toward long phrases, not a solved long phrase.

2026-05-15 08:48 PM EDT notation-renderer correction The web notation renderer now moves the treble-clef baseline upward and re-anchors flat key-signature glyphs so B-flat and E-flat render on the intended staff positions. This fixes a display-layer source of visible wrongness, but it does not count as transcription progress unless the paired audio and source-score sequence pass the strict acceptance gate.

2026-05-15 09:07 PM EDT targeted-transcription update Live v11 reprocessing exposed a workflow bottleneck: the public transcribe endpoint could only walk the stale media queue in fixed eight-sample passes, so a known evidence clip could remain stale while unrelated samples were regenerated first. The endpoint now accepts a bounded `sampleId` query parameter and an optional bounded `limit`, allowing a specific local media sample to be regenerated under the current pipeline without bypassing violin-positive checks, hidden-candidate gating, uncertainty rejection, or source-score acceptance rules. This is required for faster long-phrase iteration because failures can be retested directly instead of waiting through broad queue churn.

2026-05-15 09:20 PM EDT live targeted-reprocess verification The deployed endpoint was run against `Njh8_zq9_DM-9465`. The live tracker then reported 67.375 percent exact weighted progress, 1380 hidden transition-trace candidates, 1 accepted long-phrase counter item, 1 accepted measure-match counter item, and 1 exact-source visible match group on 2026-05-03. Browser verification of the public site showed one visible group: `Bb D C Bb D` in the score panel, local video/audio from `2:37:51-2:37:52`, and transcription `Bb4 D5 C5 Bb4 D5 / 1.1s`. Alan's later review rejected that group, so the retained progress is the targeted endpoint and hidden-candidate instrumentation, not the accepted phrase.

2026-05-15 09:24 PM EDT post-deploy state check A post-deploy API read showed that live background processing had advanced the implementation tracker to 68.125 percent exact weighted progress and 2476 hidden transition-trace candidates. That one-group state is now superseded by the 10:05 PM EDT rejection lock. The extra hidden candidates improve search coverage, but they do not count as solved transcription until a longer exact source-score run is verified and reviewed.

2026-05-15 10:05 PM EDT five-note rejection lock Alan's latest review rejected the five-note Scherzo-Tarantelle phrase itself. Curtis now demotes `A#4 D5 C5 A#4 D5` / `Bb4 D5 C5 Bb4 D5` from accepted evidence back to candidate-only material. Regression tests now require this exact lane to have zero verified score location, zero verified source snippet, no accepted measure, no accepted long phrase, no score-coordinate heat-map fragment, and no visible accepted score panel. The only items Alan had called usable remain short audio-only D/A detections; the A4/A5 score anchors, D–B-flat–G–D score sequence, and five-note Scherzo phrase are rejected until new independent review proves otherwise.

2026-05-15 10:54 PM EDT second-pass audio-gate update The repeated loop was that fast transition notes and loose reference-audio overlaps could still look like progress before the notes were independently confirmed by audio. A local audit of the rejected `Njh8_zq9_DM-9465` clip read the candidate closer to `Bb4 D5 Bb4 D5`; the displayed middle `C5` did not survive the second pass. Curtis now preserves `audioAgreement`, `agreementSources`, `agreementSourceCount`, and detector provenance in daily detected-note series. Hidden `scoreMatchCandidateNotes` are filtered before score search unless each note has audio agreement, at least one independent agreement source, spectral/onset participation, confidence at or above 0.70, and enough duration to avoid transition glitches. Symbolic score matches and reference-sequence matches now require the same per-note audio gate, and reference-audio pitch-trace matches need at least four notes and three distinct pitch classes. Regression tests now lock the rejected Scherzo phrase when the `C5` fails audio agreement, reject unagreed fast candidate notes, reject uncertain octave-corrected candidates, and stop weak two-note reference-audio coincidences from counting as phrase progress. This does not solve long-phrase transcription; it removes a false-positive lane that was making Curtis go in circles.

2026-05-15 11:17 PM EDT candidate-isolation update Curtis now separates source-verification candidates from displayable score matches in the daily-record API. Pitch-sequence candidates can still feed the truth workbench and source-verification target queue, but they no longer populate the visible `matchGroups` bucket, no longer mark a day as reference-linked, and no longer create heat-map fragments unless the exact accepted score-evidence gate passes. A new regression test proves that a locally playable Wieniawski source-sequence candidate can be queued with `candidateMatchGroups` while `matchGroups`, accepted score links, and heat-map fragments remain empty. This prevents another circular handoff where Alan has to inspect a displayed "match" and discover that it was only a candidate.

2026-05-15 11:32 PM EDT live candidate-isolation verification The deployed API reported 46.125 percent exact weighted implementation progress, 0 accepted long phrases, 0 accepted measures, 0 exact score-aligned windows, 0 actual-source score snippets, and 0 score-coordinate heat-map fragments. The 2026-05-03 daily record reported `source_verification_pending`, 0 visible `matchGroups`, 6 queued `candidateMatchGroups`, 0 score links, and 0 reference links. The top queued source target is `A D D F# D`; it remains rejected as accepted score evidence because it does not have a contiguous exact-MIDI run in the corrected source MusicXML.

2026-05-16 source-position and exact-MIDI update The verified symbolic score map itself had a staff-position error. Rechecking the local IMSLP solo-violin page at high resolution and cross-checking the Wieniawski Society bars 5–9 example showed that the opening violin entrance is `A5 G5 F5 A5 G5 F5 A5 G#5 F5`, not the prior `D6 C6 Bb5 D6 C6 Bb5 D6 C#6 B5` map. The MusicXML now stores those nine actual source notes from measures 5–7. The matcher still uses pitch-class overlap for candidate search, but accepted symbolic score matches and source-verification targets now require exact detected MIDI equality, per-note audio agreement, and the same actual-source crop and truth-evidence gates. Regression tests now cover the corrected opening map, a rejected partial exact-MIDI target whose best overlap is only `A5 G5`, and a positive audio-agreed exact-MIDI phrase `G5 F5 A5 G#5 F5` with MIDI `79 77 81 80 77`. This still does not accept a live long phrase; it removes the path that let a wrong-staff-position source map or pitch-letter-only target look solved.

5.2 Definition of solved

Long-phrase transcription is solved for Curtis only when the system can take Alan's long-form practice archive and produce evidence groups that a musician can trust without manually checking every note. The complete goal is not merely a model that prints plausible notation. The solved system must satisfy all of the following conditions.

1. **Full archive accounting.** Every practice video from the strict practice ledger is assigned to the correct practice day, has exact uploaded duration, and has a measured or explicitly pending active-practice state.
2. **Active violin time.** Total practice time means intervals where Alan is actually playing violin. Talking, tuning, setup, walking, camera adjustment, silence, scrolling, and other non-playing intervals are excluded.
3. **Local evidence playback.** Every accepted transcription snippet has a playable local video clip and audio clip in the Curtis site. Accepted evidence does not rely on a YouTube embed.
4. **Accurate note events.** The transcribed notes match the paired audio, including repeated notes, fast scalar runs, arpeggios, shifts, string crossings, rests, and obvious restarts. A fast passage must not collapse into a false repeated pitch stream such as `D D D D`.
5. **Rhythm and timing.** The system estimates onsets, durations, rests, tempo changes, slow-practice stretches, restarts, and repeated attempts. When rhythm is uncertain, the notation must mark uncertainty rather than invent a clean answer.
6. **Professional notation.** Displayed notation uses a real notation engine or a notation-quality rendering path. Treble clef placement, staff scaling, note placement, stems, beams, rests, spacing, key signature, and measure layout must look like real sheet music, not a hand-drawn mockup.
7. **Score-source truth.** A score snippet can appear only when the score-side notes are source-confirmed. Scanned images without a verified note map are not enough. A displayed A must point to an actual A in the score.

8. **Score alignment.** For known repertoire, the played note sequence must align to a specific location in the score. The match can ignore rhythm during the first pitch-class search, but accepted score display needs exact score-note identity and a stable location.

9. **Score-free exercise handling.** If Alan invents an exercise or practices a technique pattern that is not in the score, Curtis should still transcribe and group the pattern. It should not force the pattern into Scherzo-Tarantelle, Haydn, or another piece.

10. **Repeat grouping.** Repeated attempts of the same passage are grouped as attempts, loops, restarts, or slow/faster passes. The UI should not dump the same measures repeatedly.

11. **Heat maps.** Practice density, repetition density, problem density, and improvement layers appear on actual score coordinates when a score match exists, or on the generated pattern/transcription when the material is score-free.

12. **Curtis-level observations.** Observations are specific and evidence-tied: what happened, where it happened, how often it happened, whether it improved, and why the issue blocks Curtis-level readiness. Generic advice is not accepted.

13. **Compact interface.** Expanded daily entries stay compact. The useful unit is the grouped score–transcription–audio–video pair, not a large dashboard or a long explanatory panel.

14. **Regression protection.** Wrong score-note boxes, failed transcriptions, unmatched notation, stale piece labels, and non-playing practice-time credit cannot silently reappear after a later change.

5.3 Acceptance gates

The following gates define whether progress is real. If a change fails one of these gates, it is not counted as solved, even if the UI looks better.

Archive gate.

The strict practice ledger must list every video from `violin 1` forward that is either violin-numbered or date-titled as practice. Same-day videos must merge into one daily record. Each record stores source video id, title, uploaded date, inferred practice date, uploaded duration, and processing coverage.

Practice-time gate.

Active violin time is the sum of detected playing intervals, not the uploaded duration. The API must expose measured active time, checked video time, unmeasured video time, and estimate status separately. The UI label should use total practice time for active violin time and uploaded video for total source duration.

Evidence-media gate.

A transcription line is accepted only if the exact audio and video window are available locally, playable in the same Curtis page, and time-linked to the source record.

Note gate.

A displayed note must agree with the paired audio. The note label, octave where claimed, and staff position must be internally consistent. If the audio says A4, the rendered staff note must be A4, not F, G, B, or a visually adjacent note.

Score gate.

An accepted score snippet must come from the actual piece source: a verified source-score crop, verified MusicXML engraving export, or manually verified note-coordinate map for the same reference range. Generated notation may show the transcription, but it cannot appear as the accepted score snippet.

Single-note gate.

A one-note score display is allowed only as a calibration anchor, and only if both sides explicitly show the same pitch name and the score-side coordinate is verified. It must not be used as proof of a long passage.

Phrase gate.

A real phrase match requires a contiguous pitch-class sequence in the score or pattern source. Rhythm can be ignored for early matching, but accepted phrase display must keep the exact played sequence, the exact score sequence, and any omitted or uncertain notes.

Notation gate.

Hand-built notation SVG is not the long-term display path. The production path must use a professional renderer or renderer-equivalent layout with verified clef, key, staff, note, rest, spacing, and responsive behavior.

Exercise gate.

If no score match exists and the material forms a repeated exercise, the system must label it as an exercise or pattern, preserve the generated transcription, and avoid adding a fake repertoire entry.

Observation gate.

A Curtis-level observation must cite the clip, transcription event, score measure or pattern segment, issue type, repeat count, direction of change, and readiness consequence.

UI gate.

The day row stays chronological, the expanded state is compact, and only grouped evidence appears by default. Raw untrusted transcription stays hidden or goes to the transcription-run PDF with clear status.

Paper gate.

Any meaningful progress changes this section, the system-state table, and the limitation text in the same work cycle.

5.4 Required source material

The implementation cannot be completed by code alone. It needs a source bundle that makes the audio, score, and corrections checkable. The Wieniawski Scherzo-Tarantelle solo PDF is now present locally as the first source-score asset, and its first verified symbolic map covers the nine-note opening phrase only.

1. **Raw or locally captured practice media.** Curtis needs stable local access to audio/video windows from the YouTube archive or owner-provided files. Metadata alone cannot produce practice time, clips, or transcription.
2. **Source-confirmed piece names by day.** Alan can supply piece names when automatic detection is uncertain. These labels should seed matching but never become score evidence by themselves.
3. **Symbolic scores for known repertoire.** For each piece, Curtis needs MusicXML, MEI, a verified MuseScore export, or an equivalent symbolic note map with measure numbers, pitches, durations, voices, and staff positions.
4. **Score images tied to symbolic coordinates.** To show actual score snippets, the image/PDF rendering must be linked to the same symbolic note ids or verified coordinate boxes.

5. **Manual corrections.** Alan or a reviewer must be able to correct a note, octave, onset, rhythm, score location, or false match. These corrections become training and regression data.

6. **Calibration recordings.** Labeled scales, arpeggios, open strings, repeated notes, shifts, slow exercises, and fast exercises help tune pitch detection before attempting long phrases.

7. **Gold benchmark clips.** A small set of manually checked audio–transcription–score pairs is required before the app can claim improvement. The benchmark must include easy anchors, repeated notes, fast runs, arpeggios, and real Scherzo-Tarantelle fragments.

5.5 Implementation plan

Phase 0: Make the paper the tracker

Status: complete in this revision. The first required step is to make the plan durable in the Curtis paper PDF. The paper now records the definition of solved, the current baseline, the gates, the phase plan, and the update protocol. Future changes are not complete unless the paper is checked or updated.

Phase 1: Build the canonical evidence ledger

The backend needs a stricter data model for practice evidence. The current modules already include the daily-record builder, media sampler, transcription engine, symbolic-score parser, score-asset layer, reference corpus, and correction layer. The next version should make the following entities explicit:

- `source_video`: immutable source id, title, URL, uploaded time, duration, practice-date basis, and ledger inclusion basis.
- `practice_day`: date, list of source videos, uploaded duration, active-practice coverage, piece labels, and status.
- `active_interval`: source id, start, end, confidence, detector version, reason for inclusion, and reason for exclusion when withheld.
- `media_clip`: local video path, local audio path, waveform path, source interval, duration, and playable route.
- `note_event`: onset, offset, pitch class, octave, cents deviation, confidence, detector sources, uncertainty reason, and correction state.

- `score_note`: piece id, movement, measure, voice, pitch class, octave, duration, note id, and rendered coordinates.
- `match_candidate`: played note ids, score note ids or exercise-pattern ids, match score, mismatch list, and display eligibility.
- `accepted_evidence`: only the groups that pass the gates; these are the only groups the main UI can show.
- `rejected_evidence`: wrong notes, wrong score boxes, false labels, failed transcriptions, and model guesses that must not reappear.

The acceptance condition for this phase is that a wrong A-to-B or A-to-G score crop can be stored as rejected evidence and cannot be selected by the display builder again.

Phase 2: Complete the full active-practice scan

The practice-time problem is separate from transcription. Curtis should first scan the full 234 h 57 min uploaded practice ledger for violin-playing intervals with a low-cost pass. The pass should use audio energy, violin-range voiced pitch, onset density, spectral shape, and visual fallback only where necessary. It should not require full transcription.

The fast pass should walk every practice video in chronological order and store approximate active intervals. A second exact pass should refine boundaries around detected regions. The UI should show three numbers for every day: uploaded video time, checked video time, and total practice time. While scanning is partial, the estimate stays labeled as estimated. When every video has been scanned, the estimate should be replaced by measured total practice time.

Acceptance for this phase:

- every strict-ledger video has a coverage record;
- every day has uploaded duration and active-practice status;
- non-playing windows are excluded;
- active-practice total can be recomputed from stored intervals;
- the paper table is updated with checked duration, active duration, unmeasured duration, and estimate status.

Phase 3: Generate stable local evidence clips

Every accepted interval and every candidate transcription window needs local media. Curtis should generate short MP4 clips and audio clips with stable IDs derived from source video id and time range. The browser should play the local clip and audio without YouTube embedding. The audio used for transcription must be the same audio the user can play.

Acceptance for this phase:

- each accepted transcription event links to local video and local audio;
- the site can play the clip in the same window;
- source time, local file, and rendered transcription agree;
- dead local media routes fail the evidence gate rather than rendering a broken accepted group.

Phase 4: Build the verified score library

The current weak link is score-side truth. Curtis must stop relying on guessed scanned-score boxes. For Wieniawski Scherzo-Tarantelle and any future piece, the score library needs symbolic notes and rendered coordinates.

The best path is:

1. acquire a legal public-domain score image or PDF;
2. obtain or create a symbolic MusicXML/MEI/MuseScore representation;
3. render the symbolic score to image or SVG with stable note ids;
4. map each rendered note to pitch, octave, measure, and coordinates;
5. manually verify a representative set of anchors, especially A, D, G, B, fast runs, ledger lines, accidentals, and octave-displaced passages;
6. reject any score source where the note map cannot distinguish adjacent notes reliably.

Acceptance for this phase:

- a score-side A in the UI is generated from an actual score-side A;
- the rendered crop includes enough context to be legible;

- the crop does not cut off clefs, notes, beams, stems, or measure context;
- single-note anchors are visually verified before display;
- phrase snippets use the exact matched symbolic notes, not a guessed image region.

Phase 5: Replace fragile note extraction with a multi-stage transcription engine

The note detector needs to work on violin practice, not only on clean isolated notes. The engine should process active intervals in stages:

1. normalize audio and remove long silence;
2. isolate violin-dominant energy where possible;
3. detect onsets and repeated attacks before pitch smoothing;
4. run multiple pitch estimators over the same window;
5. separate stable pitch, glissando/shift, vibrato, noise, and uncertain frames;
6. group frames into note events by onset, pitch stability, and duration;
7. preserve fast pitch changes instead of smoothing them into one repeated pitch;
8. estimate rests and pauses between attempts;
9. detect double stops or chords as multi-note events when monophonic assumptions fail;
10. attach confidence and uncertainty to every event;
11. store rejected alternatives so repeated failures can be debugged.

Acceptance for this phase:

- open strings and simple scales transcribe correctly;
- repeated notes remain repeated notes with distinct onsets;
- arpeggios become changing pitch sequences, not a repeated-note collapse;
- obvious fast runs preserve the contour and most pitch classes;
- uncertain events are labeled uncertain rather than shown as clean notation;

- every displayed note can be played from its paired audio.

Phase 6: Make transcription score-aware

For known repertoire, Curtis should not transcribe in isolation and then guess the piece later. The user-supplied or detected piece name should provide a score prior. The aligner should search the score for pitch-class sequences that match the audio-derived note stream, allowing restarts, skipped notes, slow practice, repeated loops, tempo changes, and rhythm mismatch during early search.

The alignment should use multiple layers:

- pitch-class subsequence search for easy anchors;
- interval-contour search when absolute pitch is uncertain;
- dynamic time warping for variable tempo;
- Viterbi-style path selection through the symbolic score for longer passages;
- restart and loop detection when the path jumps backward;
- score-free fallback when no score path is plausible.

Acceptance for this phase:

- one-note anchors pass only when the score coordinate is verified;
- three-to-five-note sequences can be matched to exact score notes;
- longer phrases show played notes beside matched score notes;
- mismatches are visible rather than hidden;
- rhythm mismatch can be tolerated during matching but not during final notation labeling;
- accepted score matches survive a musician spot-check.

Phase 7: Support score-free technique exercises

Alan may invent an exercise during practice. Curtis must treat this as first-class evidence. If the note stream is coherent but does not match the named score, the system should cluster repeated patterns, render generated notation, and label the material as a technique pattern or exercise.

Acceptance for this phase:

- repeated made-up patterns group together;
- the generated transcription is linked to audio/video;
- no fake score snippet appears;
- the repertoire entry records technique-pattern evidence separately from piece evidence;
- heat maps can show pattern density without pretending there is a score location.

Phase 8: Add manual correction and training

Fully automatic transcription will not become reliable without corrections. Curtis needs a correction UI that allows Alan to say: this note is A, this staff note is wrong, this score box points to B, this clip is non-playing, this rhythm is wrong, this is Scherzo-Tarantelle, this is a made-up exercise, or this score match is false.

Every correction should become data:

- corrected note events become gold labels;
- rejected matches become negative examples;
- accepted matches become benchmark fixtures;
- corrected piece names update source labels without unlocking score snippets by themselves;
- corrections invalidate stale derived evidence.

Acceptance for this phase:

- Alan can fix a wrong note or score box without editing code;
- the corrected example appears in the benchmark set;
- the same failure cannot reappear after rerendering;
- the PDF tracker records the correction count and benchmark count.

Phase 9: Build the benchmark suite

Curtis needs objective tests before claiming improvement. The benchmark should include:

- open-string clips;
- G, D, A, and E major/minor scales where available;

- arpeggios;
- repeated single-note attacks;
- fast scalar runs;
- shifts with sliding or unstable pitch;
- Scherzo-Tarantelle fragments with verified score locations;
- Haydn fragments with verified score locations;
- score-free exercises;
- non-playing windows that must not count as practice time.

Metrics should include note precision, note recall, pitch-class accuracy, octave accuracy when claimed, onset error, duration error, false-note rate, repeated-note recall, phrase-match precision, phrase-match recall, score-coordinate accuracy, false accepted score match count, active-practice precision, active-practice recall, and UI evidence completeness.

Acceptance for this phase:

- tests fail on the known A-to-wrong-score-note examples;
- tests fail on repeated-pitch collapse;
- tests fail when accepted notation lacks paired audio/video;
- tests fail when practice time includes non-playing footage;
- tests pass only when the displayed evidence is actually trustworthy.

Phase 10: Use professional notation rendering

The current custom notation path is only a temporary evidence display. The solved system should render notation from MusicXML, MEI, or another structured notation representation through a professional renderer or renderer-equivalent layout. The renderer must handle treble clef placement, staff scale, ledger lines, rests, stems, beams, ties, slurs where known, key signatures, accidentals, note spacing, measure widths, and responsive crop behavior.

Acceptance for this phase:

- treble clef is correctly centered around the G-line spiral;

- the staff note for A4 appears in the A4 position;
- key signature and accidentals follow the chosen notation policy;
- snippets look publication-quality at desktop and mobile widths;
- the same notation source can export to the transcription-run PDF.

Phase 11: Build heat maps from matched coordinates

Heat maps should be computed from actual practice behavior. For score-matched passages, each matched score note or measure receives density values. For score-free exercises, each pattern segment receives density values.

The layers are:

- practice density: active time spent on the passage;
- repetition density: number of attempts or loops;
- problem density: restarts, stops, slowdowns, unstable notes, and failed attempts;
- improvement: later attempts cleaner than earlier attempts.

Acceptance for this phase:

- a heat-map mark can be traced to clips and note events;
- a score-free exercise can still have a pattern heat map;
- heat maps do not appear as decorative overlays when no match exists;
- the day-level heat map and repertoire heat map use the same evidence.

Phase 12: Generate specific Curtis-level observations

Observation generation should happen after transcription and alignment, not before. The observation engine should compare attempts of the same passage and classify issues as pitch, rhythm, bow, shift, tone, articulation, tempo, consistency, musical shape, or visible tension.

The output for each main passage should include:

- likely passage or exercise pattern;
- evidence clip;

- transcription snippet;
- score snippet when verified;
- exact issue;
- number of attempts where it occurred;
- whether it improved, stayed the same, or got worse;
- why it blocks Curtis-level readiness.

Acceptance for this phase:

- observations cite evidence;
- generic advice is absent;
- a day ends with one specific main blocker;
- rejected or uncertain transcription does not feed readiness claims.

Phase 13: Keep the UI compact

The main page should show the paper card, analyzed practice days, and repertoire. Expanded daily entries should show one or a small number of grouped evidence pairs, not a full debug dashboard. The day marker should remain visible on the left so boundaries between days are clear. Videos should be small enough that a day entry fits on screen when possible.

Acceptance for this phase:

- one daily group contains score, clip, audio, transcription, and the minimum useful status;
- raw text explanations are removed from the main view;
- chronological order is stable;
- desktop and mobile render without overlap, excessive height, or broken media controls;
- unresolved material is visible as state, not as a fake dashboard.

Phase 14: Automate paper updates

The paper is now the progress tracker. Future meaningful Curtis changes should update:

- live state numbers;
- solved gates;
- benchmark metrics;
- accepted score-match count;
- accepted transcription duration;
- full archive scan coverage;
- remaining blockers;
- limitations.

Acceptance for this phase is procedural: after every Curtis change that affects transcription, score matching, practice time, clips, heat maps, repertoire, observations, or evidence gates, the LaTeX paper is checked, rebuilt, and the root `paper.pdf` is refreshed.

5.6 Milestones

5.7 Immediate next implementation

After this PDF tracker, the first concrete implementation target is **full active-practice coverage plus benchmark/correction scaffolding**. This is the first useful code milestone because transcription cannot be trusted without knowing which source windows contain violin playing, and it cannot improve reliably without a gold correction loop.

The 2026-05-13 11:57 AM EDT backend update completed the coverage and correction scaffolding portion:

- added a canonical active-practice coverage ledger exposed through the review API and `/api/curtis/active-practice-coverage`;
- separated uploaded video duration, checked sampled video duration, measured active practice time, candidate violin windows, unmeasured video time, and estimate state;

Table 4. Milestone plan for moving from current evidence gates to solved long-phrase transcription.

Milestone	Target	Acceptance evidence	Status
M0	Paper tracker	Roadmap is included in the Curtis paper PDF and becomes the update record.	Complete in this revision.
M1	Practice time	All 234 h 57 min of strict-ledger video has active-practice coverage and per-day totals.	Active-scan route deployed; live state now has 331 active intervals from 106 sample results, 2 h 28 min checked video, 2 h 21 min measured active practice, and 250 queued windows; full chronological scan still pending.
M2	Score truth	Scherzo-Tarantelle has verified symbolic notes and score coordinates.	Partially advanced; the local source PDF is present and a nine-note symbolic opening phrase is now parsed from source-backed MusicXML. More measures and score coordinates remain pending.
M3	One-note anchor	A displayed A from audio is matched to a verified score-side A, with no wrong-note crop.	Not accepted yet; audio-only D/A fragments are useful, but all A-specific score anchors remain rejected until the score-side notehead and octave are verified.
M4	Short sequence	Three-to-five played notes align to exact score notes and render with local audio/video.	Reopened after review; the five-note Scherzo phrase is rejected, and new phrase candidates must pass the second-pass audio gate plus source-score crop review.
M5	Long phrase	A 10–30 s passage transcription matches the paired audio and score location.	Pending.
M6	Repeat grouping	Multiple attempts of the same passage group into loops, restarts, slow attempts, and faster attempts.	Pending.
M7	Heat map	Practice density and problem density appear on verified score coordinates.	Pending.
M8	Repertoire update	Piece entries update from accepted daily evidence and show active time, clips, snippets, and blockers.	Pending.
M9	Readiness observation	Main blocker is generated from accepted evidence, not generic advice.	Pending.
M10	Regression lock	Tests prevent failed notation, wrong score boxes, and non-playing practice-time credit from returning.	Correction benchmark scaffold added; user-reviewed wrong-score regressions are stored; truth-workbench write gates reject accepted audio-score note mismatches; visible-note-sequence verification is required for accepted score panels; second-pass audio agreement now blocks weak fast-transition and reference-audio phrase candidates.

- counted checked non-playing sampled windows as checked video without adding them to practice time;
- added `/api/curtis/evidence-corrections` for note, score, match, piece, and active-interval corrections;
- added benchmark fixtures for rejected score-note and match corrections;
- added regression tests for checked non-playing windows, wrong score-note benchmark creation, and rejecting accepted score evidence when the displayed score note conflicts with the observed note.

The 2026-05-13 12:33 PM EDT backend update completed the active-scan route portion:

- added a persistent `activePracticeScan` state record with active intervals, sample results, pending windows, and run history;
- scans stored local media samples only after the violin-presence gate is positive;
- extracts sound-active sub-intervals inside the source window and stores them as active-practice intervals;
- records negative violin-presence samples as checked without adding practice time;
- feeds persisted active intervals into per-day totals and the active-practice coverage ledger;
- exposes active-scan status and run endpoints so the deployed runtime can perform the pass without transcription;
- queues the next unchecked 90-second strict-ledger windows for owner media capture.

The 2026-05-13 12:59 PM EDT deployed scan update completed the first live run and exposed a practice-time accounting failure mode: sparse, captured samples can make an archive-wide estimate look too certain. The implementation now records active-scan status counts, caps active seconds by checked source coverage, and treats active-scan results as the practice-time source for scanned samples instead of letting noisy transcription durations override them.

The 2026-05-13 03:26 PM EDT queue update connected the active-scan queue to the owner-media sync path. The sync path now asks the deployed active-scan run endpoint for pending windows, keeps the response compact, and places those windows ahead of generic expansion or blind sampling. The pending queue remains the source of work for M1: each captured window must still be downloaded or browser-captured, locally classified for violin presence, uploaded, rescanned, and recorded before it becomes checked practice-time coverage.

The 2026-05-13 03:37 PM EDT owner-sync run proved that the local upload/rescan loop still works after the queue update. It uploaded one cached violin-positive 5/3 window and the deployed active scan converted it into one additional active-practice interval. Because cached violin-positive media is still prioritized ahead of new capture, this run did not consume the first chronological pending window. The next M1 step should either exhaust the cached positive backlog first or run a queue-only owner-sync mode so the pending list starting at `violin 1` advances.

The 2026-05-13 04:05 PM EDT update added that queue-only owner-sync mode and used it to consume the first chronological pending window. The `violin 1 *0-90` window was captured, classified as non-playing/unclear, uploaded, and folded into the scan results as checked non-violin evidence. This is correct progress because it prevents setup or non-playing footage from being counted as violin practice. It does not advance transcription because no violin-positive musical material was present in that window. The UI now shows the current long-phrase transcription completion percentage and the complete done/open gate list at the bottom of the Curtis page so the remaining work is visible without turning daily entries into a large debug dashboard.

The 2026-05-13 05:44 PM EDT update advanced the same queue from `*90-180` through `*450-540`. Those windows produced active-practice evidence and moved checked coverage to 2 h 3 min. The timeout exposed a resume failure: cached active-scan pending windows could be captured but then skipped on the next queue-only run. The helper now includes matching cached pending-window files in queue-only order, including non-playing cached windows, so an interrupted scan can resume rather than forcing Alan or Codex to manage the cache manually. The coverage ledger also now treats unchecked transcription-only remnants as zero practice time; active practice cannot exceed checked video coverage.

The 2026-05-13 06:01 PM EDT update consumed the next chronological queue window, `*540-630`. That window was violin-positive and added four active intervals after the active-practice scan. The run succeeded without a timeout at batch size one, and the next pending window became `*630-720`. This kept the current work focused on M1 coverage rather than score matching or long-phrase notation.

The 2026-05-13 06:26 PM EDT update consumed the next chronological queue window, `*630-720`. That window was violin-positive and added five active intervals after the active-practice scan. The command wrapper timed out while the owner-sync process was still running, so the process was allowed to finish and then verified from live state instead of starting a duplicate run. The next pending window is now `*720-810`.

The 2026-05-13 07:55 PM EDT update consumed the next chronological queue window, `*720-810`. That window was violin-positive and added three active intervals after the active-practice scan. The owner-sync process completed cleanly with a longer timeout, so no duplicate capture or manual cache recovery was needed. The next pending window is now `*810-900`.

The 2026-05-13 08:41 PM EDT status-surface update made the Curtis page itself carry the implementation state instead of leaving the percentage and plan in the chat thread. The bottom tracker is now labeled as long-phrase transcription implementation status and is driven by the same backend gate object as the API. The visible panel reports 29 percent completion, 2 h 7 min checked video out of 213 h 23 min, 2 h measured active practice, 0 exact score windows, 0 accepted long phrases, 250 queued windows, and 250 active intervals. It also exposes a compact done list, an open-work list, an eight-phase implementation plan, and a collapsed detailed gate checklist. This is a presentation and tracking improvement, not a transcription breakthrough: score-truth verification, phrase-level note/rhythm extraction, exact score alignment, heat maps, and Curtis-level observations remain open.

The 2026-05-13 09:25 PM EDT implementation pass advanced the M1 scan and M10 benchmark loop. The queue-only sync consumed `violin 1` windows `*810-900` and `*900-990`; both became violin-positive source evidence and the deployed active-practice scan folded them into the checked ledger. Live state now reports 31 percent implementation completion, 2 h 10 min checked video out of 213 h 23 min, 2 h 3 min measured active practice, 92 sample results, 82 active-violin samples, 10 checked non-violin samples, 257 active intervals, and 250 queued windows. The correction ledger now contains three benchmark fixtures for real wrong-score-note failures: `A4/B`-crop, `A4/G4`-crop, and `A5/G5`-crop. Those cases are progress only because they prevent the same bad score matches from being counted as solved evidence. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-13 10:05 PM EDT implementation pass continued the M1 scan and fixed a tracker presentation gap. The queue-only sync consumed `violin 1` windows `*990-1080` and `*1080-1170`; both became violin-positive source evidence and the deployed active-practice scan folded them into the checked ledger. Live state now reports 31.13 percent exact weighted implementation completion, 2 h 13 min checked video out of 213 h 23 min, 2 h 6 min measured active practice, 94 sample results, 84 active-violin samples, 10 checked non-violin samples, 269 active intervals, and 250 queued windows. The rounded public label still reads 31 percent, but the API and site now expose exact weighted progress so small M1 coverage gains do not disappear. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-13 11:14 PM EDT implementation pass continued M1 and tightened the progress calculation itself. The queue-only sync consumed `violin 1` windows `*1170-1260` and `*1260-1350`; the first became violin-positive source evidence and the second became checked non-playing/unclear evidence. The deployed active-practice scan folded both into the coverage ledger. Live state now reports 96 sample results, 85 active-violin samples, 11 checked non-violin samples, 283 active intervals, 2 h 16 min checked video, 2 h 7 min measured active practice, and a 199 h 19 min provisional archive estimate. The tracker implementation now sums unrounded gate points before formatting the public label, so this small but real coverage gain moves the expected exact implementation label to 31.128 percent after deployment. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-13 11:38 PM EDT implementation pass continued M1. The queue-only sync consumed `violin 1` windows `*1350-1440` and `*1440-1530`; both became violin-positive source evidence and the deployed active-practice scan folded them into the checked ledger. Live state now reports 31.131 percent exact weighted implementation completion, 2 h 19 min checked video out of 213 h 23 min, 2 h 10 min measured active practice, 98 sample results, 87 active-violin samples, 11 checked non-violin samples, 294 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-13 11:55 PM EDT implementation pass continued M1. The queue-only sync consumed `violin 1` windows `*1530-1620` and `*1620-1710`; the first became checked non-playing/unclear evidence and the second became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.134 percent exact weighted implementation completion, 2 h 22 min checked video out of 213 h 23 min, 2 h 12 min measured active practice, 100 sample results, 88 active-violin samples, 12 checked non-violin samples, 300 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 12:09 AM EDT implementation pass continued M1. The queue-only sync consumed `violin 1` windows `*1710-1800` and `*1800-1890`; both became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.136

percent exact weighted implementation completion, 2 h 25 min checked video out of 213 h 23 min, 2 h 15 min measured active practice, 102 sample results, 90 active-violin samples, 12 checked non-violin samples, 313 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 12:35 AM EDT implementation pass continued M1. The queue-only sync consumed `violin 1` windows `*1890-1980` and `*1980-2070`; both became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.139 percent exact weighted implementation completion, 2 h 28 min checked video out of 213 h 23 min, 2 h 18 min measured active practice, 104 sample results, 92 active-violin samples, 12 checked non-violin samples, 322 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 03:00 AM EDT implementation pass continued M1. The queue-only sync consumed `violin 1` windows `*2070-2160` and `*2160-2250`; both became violin-positive source evidence. The deployed active-practice scan folded both into the checked ledger. Live state now reports 31.142 percent exact weighted implementation completion, 2 h 31 min checked video out of 213 h 23 min, 2 h 21 min measured active practice, 106 sample results, 94 active-violin samples, 12 checked non-violin samples, 331 active intervals, and 250 queued windows. This is real practice-time coverage progress only. Curtis still has 0 accepted exact score windows and 0 accepted long phrases.

The 2026-05-14 05:11 AM EDT source-score acceleration pass advanced M2 and M4 instead of continuing only the slow archive scan. The implementation added a local MusicXML source excerpt for the Scherzo-Tarantelle solo-violin opening motif, wired it into the Wieniawski score target, and changed the symbolic matcher so this source target can accept a four-note measure-level run with at least three distinct pitch classes. The parser now preserves written flats such as B-flat for staff placement while still matching their pitch class to detected audio, which closes the failure mode where a normalized A-sharp could render on the wrong staff position. The score snippet renderer now receives the target key signature and renders a G-minor two-flat key signature before the matched notes. The completion tracker now separates `acceptedMeasureMatchCount` from `longPhraseAcceptedCount`, so Curtis can show one verified measure-level score/audio match without claiming solved long-phrase transcription. The regression suite increased to 103 tests and now covers source-backed Wieniawski symbolic notes, four-note source-measure matching, written-flat staff placement, accepted-measure counting, and the continued rejection of one-note visual crops. This pass does not solve long-phrase transcription; it creates the first narrow, test-covered lane from source score notes to a grouped score–audio–video–transcription match.

The remaining M1 work is continuing the low-cost media pass over every unmeasured strict-ledger video in chronological windows, then replacing the sample-biased estimate with full-archive checked intervals. The remaining M10 work is the larger benchmark suite for repeated notes, arpeggios, fast runs, score-free exercises, score-coordinate truth, accepted notation/media completeness, and practice-time accounting regressions.

The long-phrase track should now run in parallel with full-archive scanning. Waiting for all 234 h 57 min of archive coverage before attempting a single measure would be too slow. The faster strategy is a narrow accepted-measure lane:

1. select one active violin-positive window from an already captured practice day, starting with 2026-05-03 because it already has hidden detected note series;
2. obtain or manually verify the corresponding symbolic Scherzo-Tarantelle measure from a source-backed score;
3. run the existing pitch-class phrase matcher against the stored detected series;
4. accept the group only if the exact score notes, displayed transcription, local audio, and local video all agree;
5. then extend the same source-backed score map outward to adjacent measures and longer phrases.

The next coding pass should:

1. verify one symbolic Scherzo-Tarantelle measure with exact pitch names and measure context;
2. feed that measure into the existing symbolic-score target path;
3. mine the stored 2026-05-03 detected note series for a contiguous score/audio match;
4. accept at most one compact grouped score–clip–audio–transcription pair if the evidence gates pass;
5. keep running queue-only owner-media sync in parallel, starting with `violin 1` at `*2250-2340`;

6. rerun the active-practice scan after those uploads and record checked, active, and pending totals.

Curtis can now pursue the first verified measure before M1 is complete, but it still cannot claim a score match from guessed scanned crops, one-note anchors, or reference-audio pitch traces. The speed-up is to shrink the musical target to one verified measure and make that one measure pass the same gates that a future long phrase will need.

6 Materials and methods

Backend

The backend is a FastAPI service deployed on Railway. It exposes `https://curtis.aolabs.io/api/curtis/media-status` for compact state, `https://curtis.aolabs.io/api/curtis/ops-check` for fuller operational state, local stored sample media for embedded playback after the media is violin-positive, sample indexes for duplicate-window avoidance, and run endpoints for scans, media probing, analysis, and coaching. Persistent runtime state stores sources, inventory, review findings, media samples, analysis runs, and scan history.

Source scan

The default public source is `https://www.youtube.com/@nalalan`. The scanner queries public video metadata, classifies videos by title, duration, and candidate reasons, and stores inventory records. Dated practice logs and long-form videos are treated as practice candidates. The media probe now keeps one early baseline window but then probes deeper quarter, midpoint, five-eighths, three-quarter, and tail windows before spending additional passes on early setup probes, and each stored sample is keyed by video identifier plus window start so later passes can expand coverage without overwriting earlier evidence. When configured above the dense-sampling threshold, the probe and owner-media helper generate dense 90-second windows from the start of each video through the latest valid window so the full practice archive can be checked without the earlier retention cap. Each sample is then classified by the violin-presence sampler before it can count as active practice evidence. The owner-media helper also checks its local cache of already captured browser samples, reuses classifier results when the file signature and classifier version match, prioritizes active-scan pending windows before generic expansion or blind sampling, can run in queue-only mode to consume chronological pending windows without falling back to unrelated cached positives or blind windows, resumes cached queue windows after an interrupted capture, includes cached negative queue windows so non-playing time is checked rather than ignored, prioritizes unsynced violin-positive cached windows during normal mode before capturing more blind windows, and expands around already-found violin-positive windows before returning to blind sampling. The practice-hours total uses a stricter title-confirmed ledger from `violin` 1 onward: violin-numbered videos and date-titled logs count toward the top-line session duration, while unrelated long videos remain indexed but outside the practice-hours total. The scan does not itself inspect performance quality.

Model review

The model layer reviews sampled media sections where usable excerpts exist. GPT-5.5 is used for text and visual review, and separate audio models are configured for audio review and piece-title verification. Findings are sanitized so weak evidence terms such as obscured, not audible, or no clear evidence produce an unjudged state rather than an unsupported critique. Piece labels are also sanitized: coach reviews can store candidate evidence but cannot create confirmed repertoire titles, and the piece-identification layer must provide source-window evidence plus independent verification before it updates the current piece list. Same-video consensus can compare multiple sampled windows before accepting a title; for long date-stamped practice captures with later samples, setup or non-playing windows are excluded from confirmation, and confirmed active titles are versioned so older labels are rechecked when source rules change. User-rejected false labels are excluded from later passes only for the source where the correction applies, so later videos can still identify that repertoire when present. A source correction also invalidates older confirmed labels for that same source instead of promoting the next stale candidate, and repeated corrected false labels keep that source out of the confirmed repertoire list until a stronger acceptance path exists. Rejected proposed titles, top candidates, and verification titles are scrubbed from current API output, not only from confirmed piece records. The 5/1 forensic pass rejected prior Paganini, Bruch, Mendelssohn, Brahms, Sibelius, Tchaikovsky, Wieniawski, Saint-Saens, Ravel, Bazzini, Ernst, Carmen, Zigeunerweisen, Bach Partitas, Kreisler Praeludium and Allegro, Sarasate family labels, Ysaye Sonata No. 3 Ballade, Mozart K. 216, Glinka Ruslan and Lyudmila, Strauss Till Eulenspiegel, Strauss Don Juan, Smetana Bartered Bride, Prokofiev Classical Symphony, Dvorak Carnival Overture, Shostakovich Symphony No. 5, Ravel Bolero, the first broad orchestral candidate batch, and the later overture, dance, and pops-orchestral candidate batch for that source. Alan then supplied the ground-truth source label: Haydn Symphony No. 94, IV.

Finale, Violin I part. The system now treats that label as human-confirmed for 5/1, stops injecting new seeded guesses for the same source, and keeps Curtis-level completion unscored until judged playing evidence supports a score. Alan also supplied the 5/2 source label, Wieniawski Scherzo-Tarantelle, Op. 16, and then confirmed that all violin footage on 5/3 was the same piece. The training state now records all three source labels as confirmed sources and reports zero score-aligned windows until pitch/rhythm extraction has matched a source window to a score section. Transcribed samples can now produce learned pitch/rhythm fingerprints only after their media window is violin-positive, and visible notation requires note/rhythm verification. Future samples can receive a nearest-source hint when their extracted notes and rhythms resemble a prior confirmed source, an explicit title-labeled calibration upload, a symbolic reference, or a public labeled YouTube reference indexed through the official Data API; if the extracted pattern does not belong to a known score, it remains piece-or-exercise pending rather than being forced into repertoire. Public reference metadata can improve candidate vocabulary and target selection, but it does not become an audio fingerprint until a permitted media source exists. Possible labels and model-only verified titles remain uncertain daily evidence unless a source label is confirmed, so future misses appear as missing evidence rather than as wrong repertoire. Piece-identification output does not create a Curtis-level readiness percentage by itself; readiness scoring requires judged playing evidence. Dated practice entries are keyed by the source-video date rather than the later processing time, so historical uploads cannot masquerade as today's practice state. The owner-browser sync uses the sample index to prioritize uncaptured public YouTube windows rather than re-uploading already reviewed excerpts, and the piece-identification pass can test multiple active moments inside each captured sample before withholding the title. The daily-record builder joins the strict `violin 1` ledger to violin-positive media samples, hidden machine note/rhythm events, practice-time evidence, clips, score packet support, repeated-fragment or repeated-pattern heat maps, score-aware read fields, and observation records. Repertoire entries are built only from confirmed daily-record evidence and keep progress percentages unassigned until clips, notation, and score or pattern alignment support them.

Progress ledger

AO Progress tracks Curtis as part of a larger multi-application state record. The first progress snapshot included Curtis home, Curtis media state, other AO Labs apps, Imagineer state, Imagineer paper, Relay state, and public project surfaces. Curtis contributes dated practice inventory, current focus, review status, and future trend signals.

7 Limitations

The current system cannot predict acceptance at Curtis. It also cannot replace a musician, teacher, jury, studio lesson, or audition process. It can be wrong when the source media are incomplete, when audio is unavailable, when video frames hide the physical action, or when model review fails. The broad micro-transcription gate is currently held closed: candidate note runs can be stored, but no displayed passage transcription should be treated as solved until its notes audibly match the paired clip. The only current displayed notation is two strict audio-checked note fragments, so the useful claim is limited: Curtis Media Review turns practice media into a persistent, source-bounded record that can support iteration over time.

References

1. K. A. Ericsson, R. T. Krampe, C. Tesch-Romer, The role of deliberate practice in the acquisition of expert performance. *Psychological Review* **100**, 363–406 (1993).
2. A. Williamon, Ed., *Musical Excellence: Strategies and Techniques to Enhance Performance* (Oxford University Press, Oxford, 2004).
3. G. Klickstein, *The Musician's Way: A Guide to Practice, Performance, and Wellness* (Oxford University Press, New York, 2009).
4. A. N. Pham, *Curtis Media Review*, Live software system, curtis.aolabs.io, 2026.
5. J. P. Bello *et al.*, A tutorial on onset detection in music signals. *IEEE Transactions on Speech and Audio Processing* **13**, 1035–1047 (2005).
6. A. de Cheveigne, H. Kawahara, YIN, a fundamental frequency estimator for speech and music. *The Journal of the Acoustical Society of America* **111**, 1917–1930 (2002).

7. E. Benetos, S. Dixon, D. Giannoulis, H. Kirchhoff, A. Klapuri, Automatic music transcription: challenges and future directions. *Journal of Intelligent Information Systems* **41**, 407–434 (2013).

8. M. Muller, *Fundamentals of Music Processing: Audio, Analysis, Algorithms, Applications* (Springer International Publishing, Cham, 2015).

9. E. J. Humphrey, J. P. Bello, Y. LeCun, presented at the Proceedings of the 13th International Society for Music Information Retrieval Conference, pp. 403–408.

10. A. N. Pham, *AO Progress Ledger*, Live longitudinal state system, progress.aolabs.io, 2026.

11. Google for Developers, *YouTube Data API* (2026; `https://developers.google.com/youtube/v3`).

12. W3C Music Notation Community Group, *MusicXML* (2026; `https://www.musicxml.com/`).

13. International Music Score Library Project, *IMSLP: Petrucci Music Library* (2026; `https://imslp.org/`).